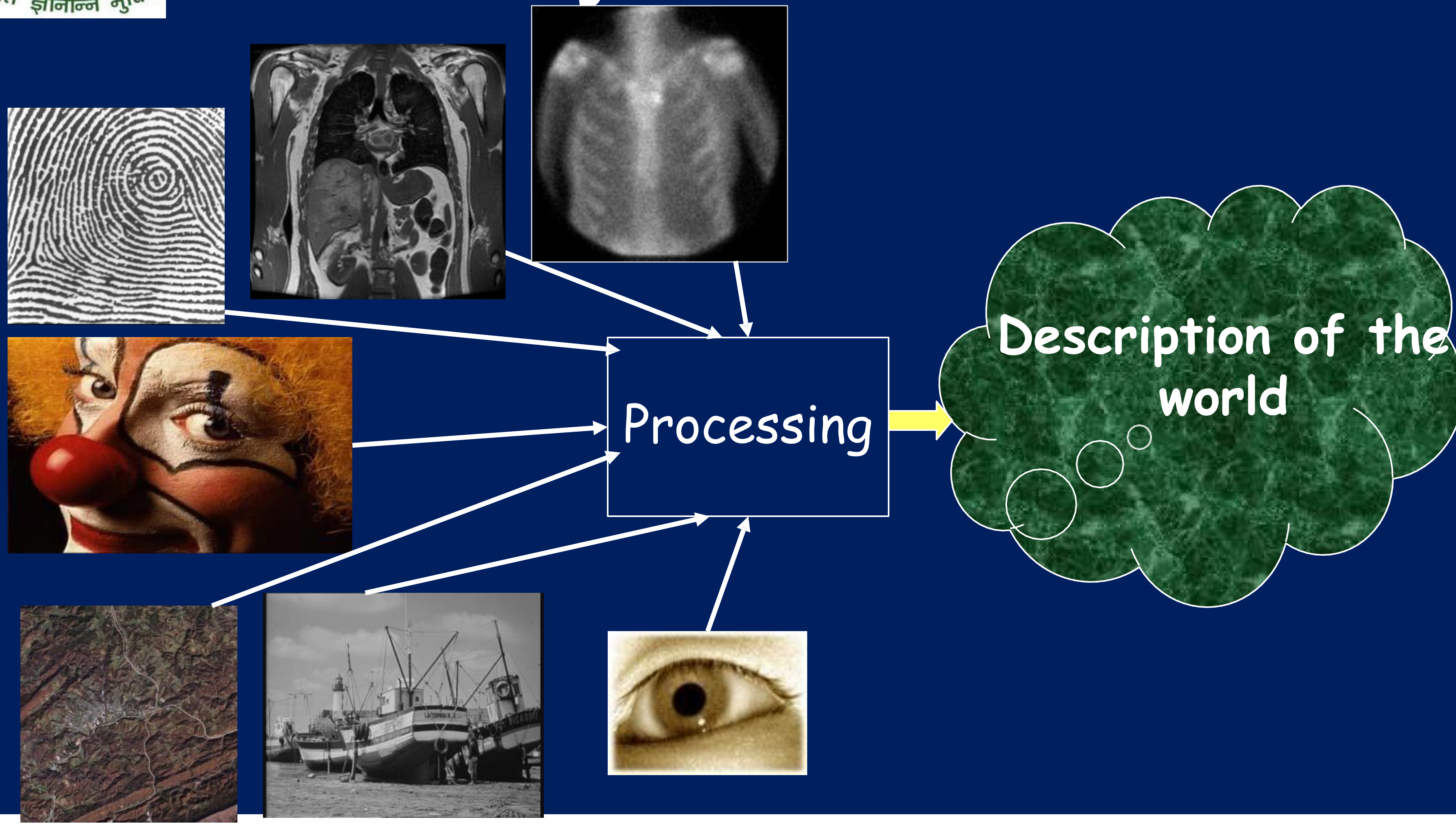
MCSC0009: Image Processing And Analysis



MCS 2004 General Information

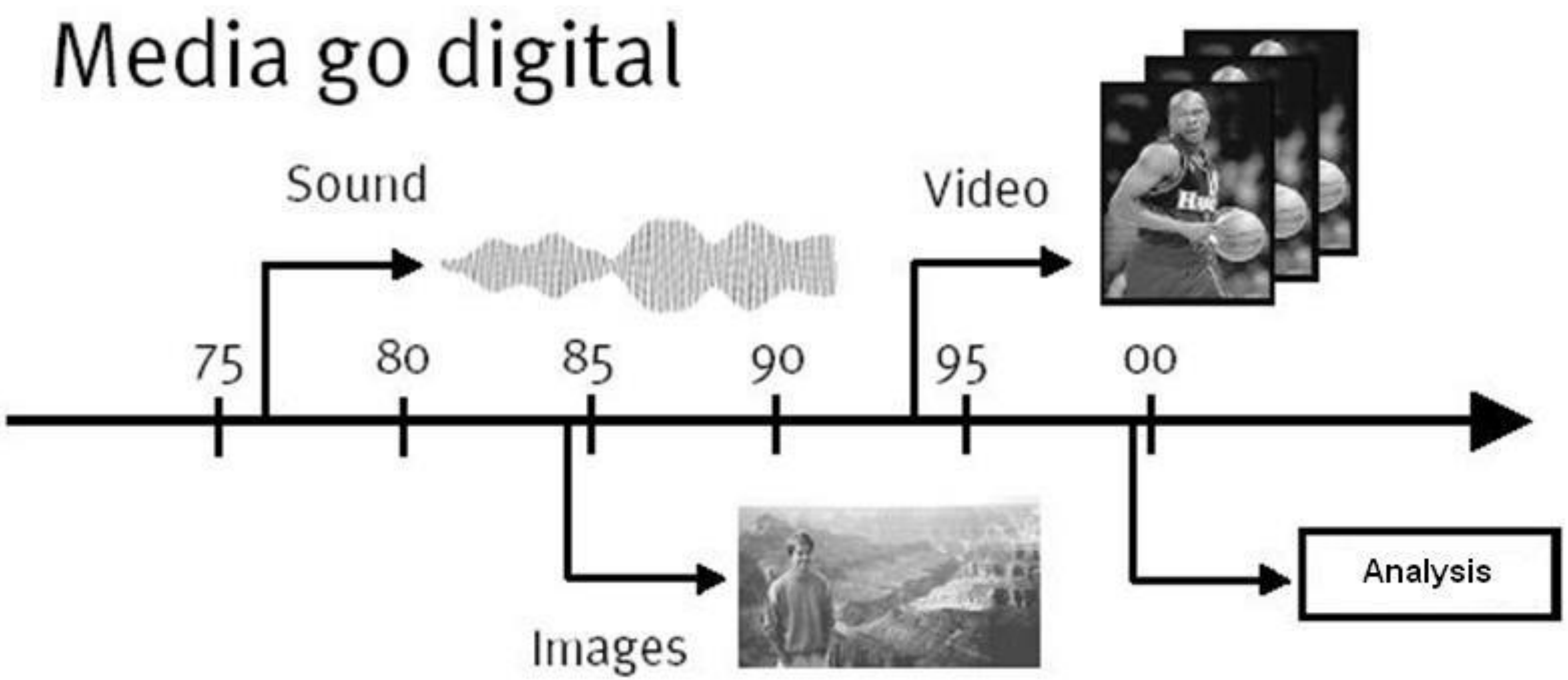
- **Suggested textbook:**
  - R.C. Gonzalez and R.E. Woods, "**Digital Image Processing**", 3rd edition, Prentice-Hall' 2011
  - Chanda, Bhabatosh, Majumder, Dwijesh Dutta "**Digital Image Processing And Analysis**" 2nd edition, PHI Learning

- **Prerequisites**
  - Knowledge of the following three areas:
    - Linear algebra, Elementary probability theory, Digital

Teaching Objectives

- **By the end of this semester, you will**
  - Know basics of Digital Image Processing and Computer Vision including Image Acquisition, Perception, Transformation, Enhancement, Compression, Segmentation, Analysis, and so on .....
  - Be able to use MATLAB to implement basic image processing algorithms and get familiar with some functions provided by MATLAB image processing toolbox

Medi
a



More than 80% of information is received by visual perception

Media …

- Early days of computing, data was numerical.
- Later, textual data became more common.
- Today, many other forms of data exist: voice, music, speech, images, computer graphics, etc
- Each of these types of data are signals.
- Loosely defined, a signal is a function that conveys information.

A Picture in worth a

1,000 words

Digital Image Processing

- A major portion of the information received by a human from the environment is visual

- Processing visual information by computer is drawing a lot of attention by the researchers.

- The process of receiving & analyzing (processing) visual information by digital computer is called **Digital Image Processing**

What is a Digital Image?

- A **digital image** is a representation of a two-dimensional image as a finite set of digital values, called picture elements or pixels

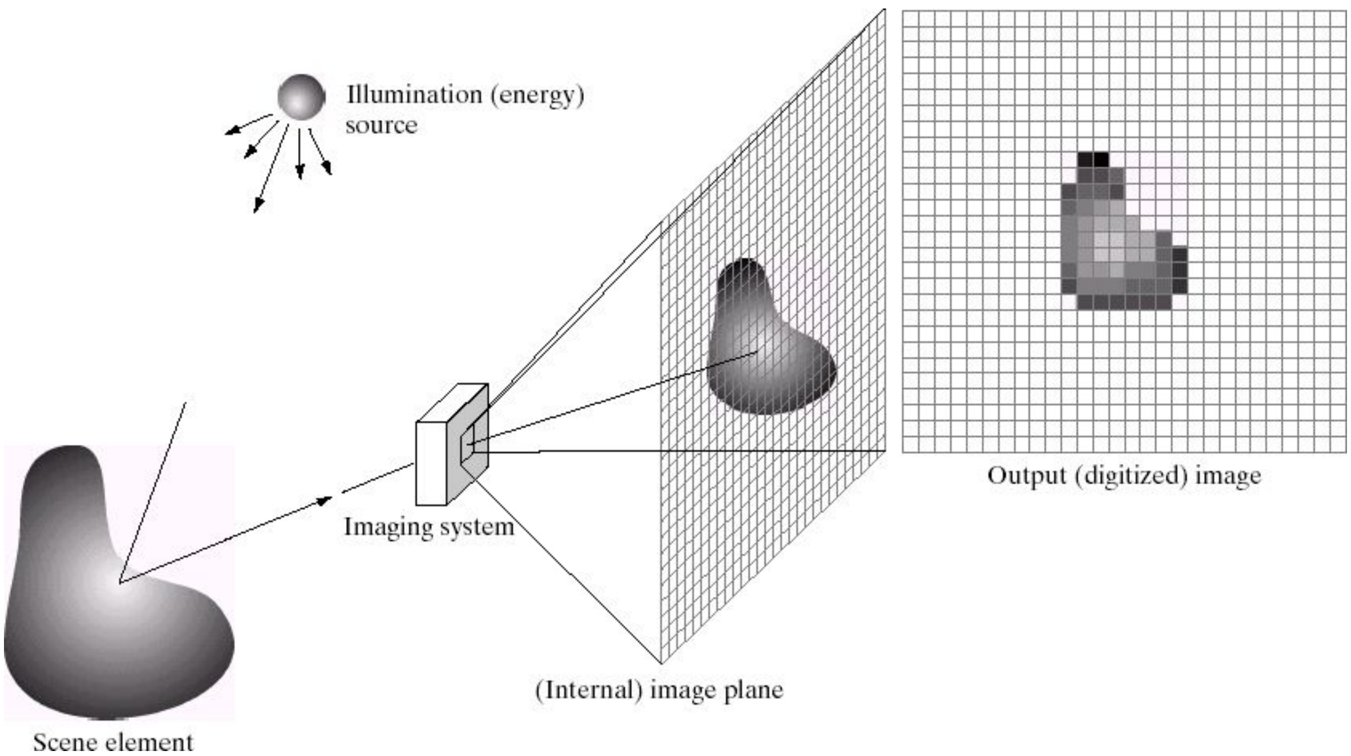
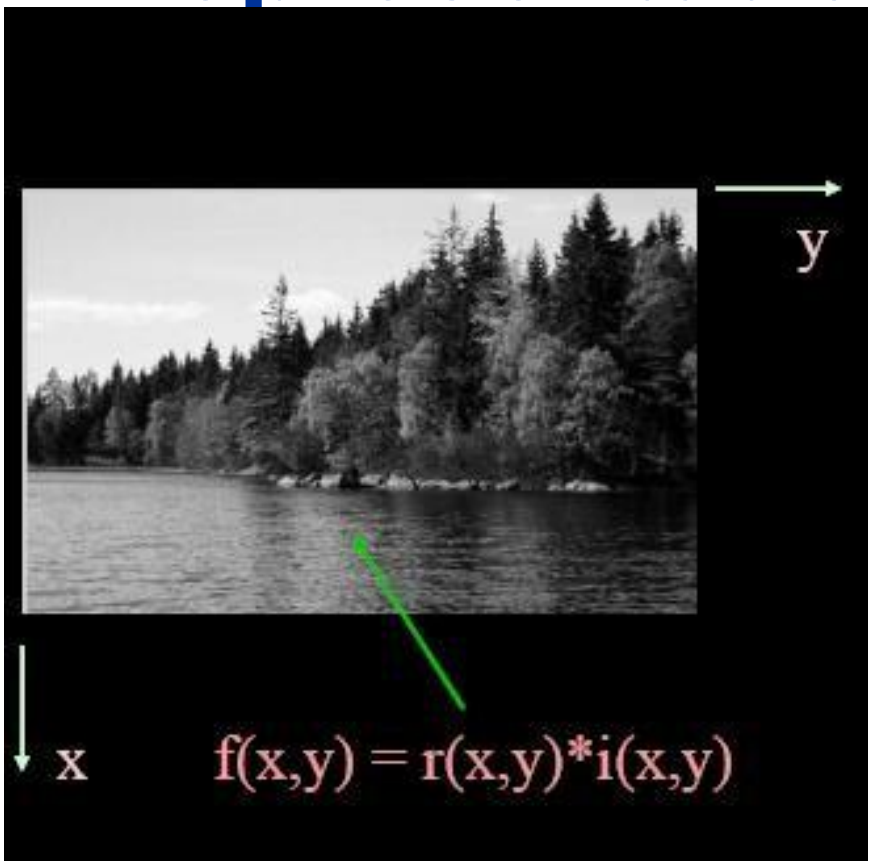


Image Representation



r(x,y) – reflectance of surface (0-1) i(x,y) – intensity of light (0-infinite)

- An image is a 2-D light intensity function f(x,y)
- A digital image f(x,y) is discretized both in spatial coordinates and brightness
- It can be considered as a matrix whose row, column indices specify a point in the image and the element value identifies gray level value at that point
- These elements are referred to as pixels or pels

Image Representation …

An image is a two-dimensional function **f(x,y)**, where x and y are the **spatial** (plane) coordinates, and the amplitude of **f** at any pair of coordinates (x,y) is called the intensity of the image at that level.

If x,y and the amplitude values of f are finite and discrete quantities, we call the image a digital image. A digital image is composed of a finite number of elements called pixels, each of which has a particular location and value.

Image Representation …

- Spatial discretization by grids
- Intensity discretization by quantization

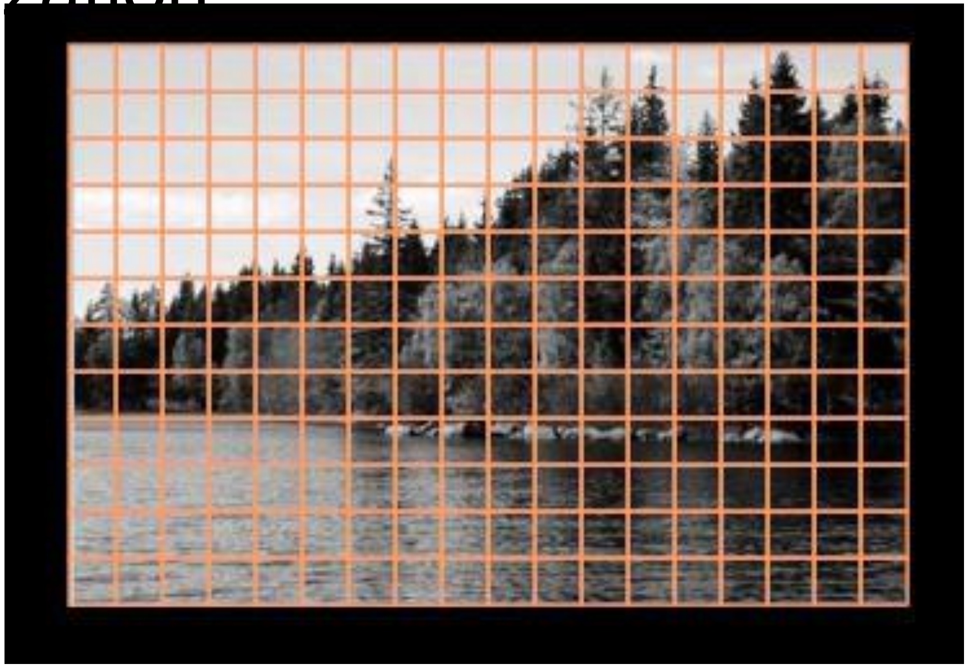


Image Representation …

$$I = \begin{bmatrix} f(0,0) & f(0,1) & f(0,2)\ldots\ldots\ldots f(0,N-1) \\ f(1,0) & f(1,1) & f(1,2)\ldots\ldots\ldots f(1,N-1) \\ f(2,0) & f(2,1) & f(2,2)\ldots\ldots\ldots f(2,N-1) \\ \vdots & \vdots & \vdots \qquad\qquad \vdots \\ f(M-1,0) & f(M,1) & f(M,2)..f(M-1,N-1) \end{bmatrix}$$

- Image Size : 256x256, 512x512, 1024x1024 etc
- Quantization: 8 bits

Image Representation …

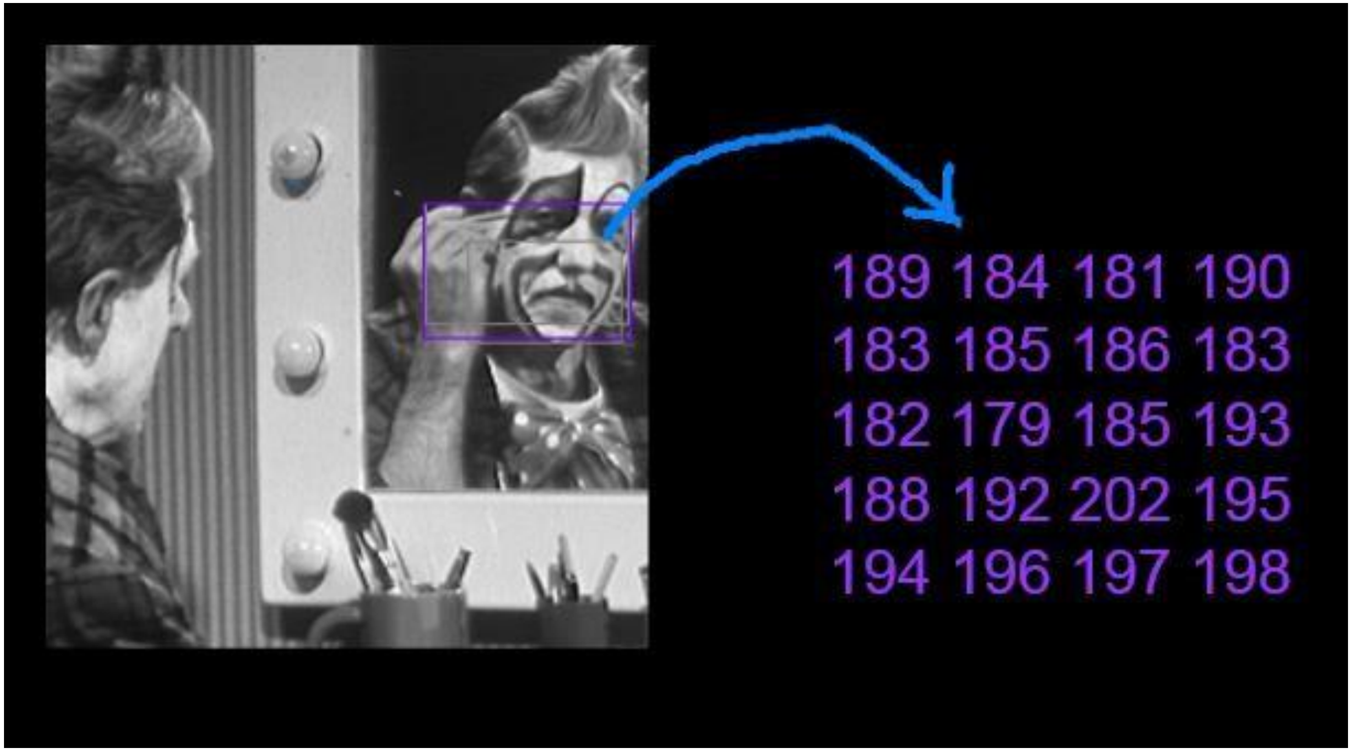


Image Representation …

Pixel intensity value

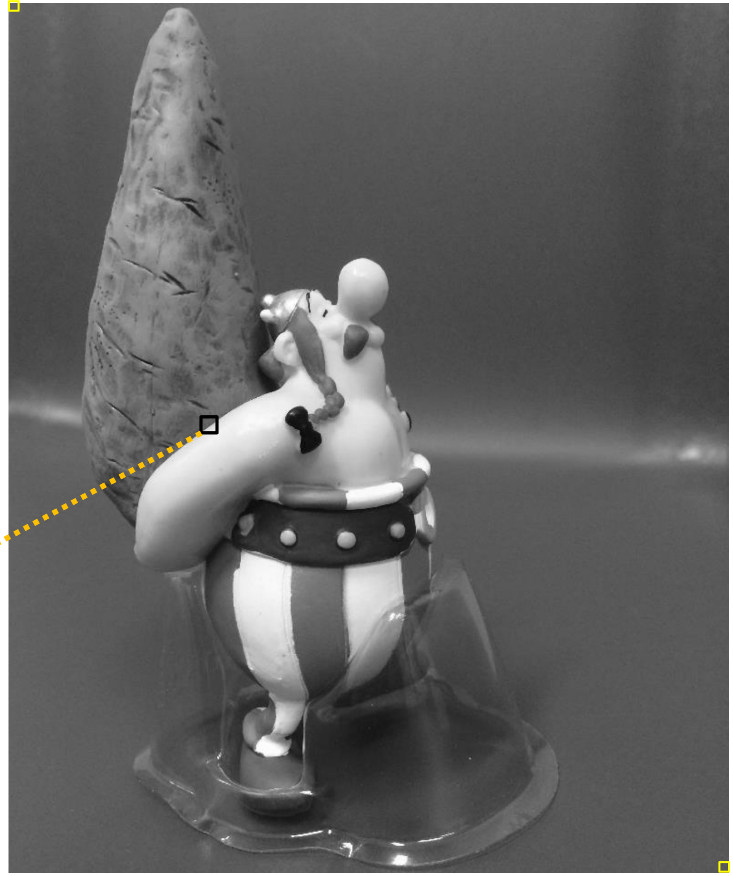
f(1,1) = 103

Pixel location

rows columns

f(645:650,1323:1328) =

83	82	82	82	82	82
82	82	82	81	81	81
82	82	81	81	80	80
82	82	81	80	80	79
80	79	78	77	77	77
80	79	78	78	77	77



f(2724,2336) = 88

Consider the following image (2724x2336 pixels) to be 2D function or a matrix with rows and columns

In 8-bit representation Pixel intensity values change between 0 (Black) and 255 (White)

Digital Image Processing …

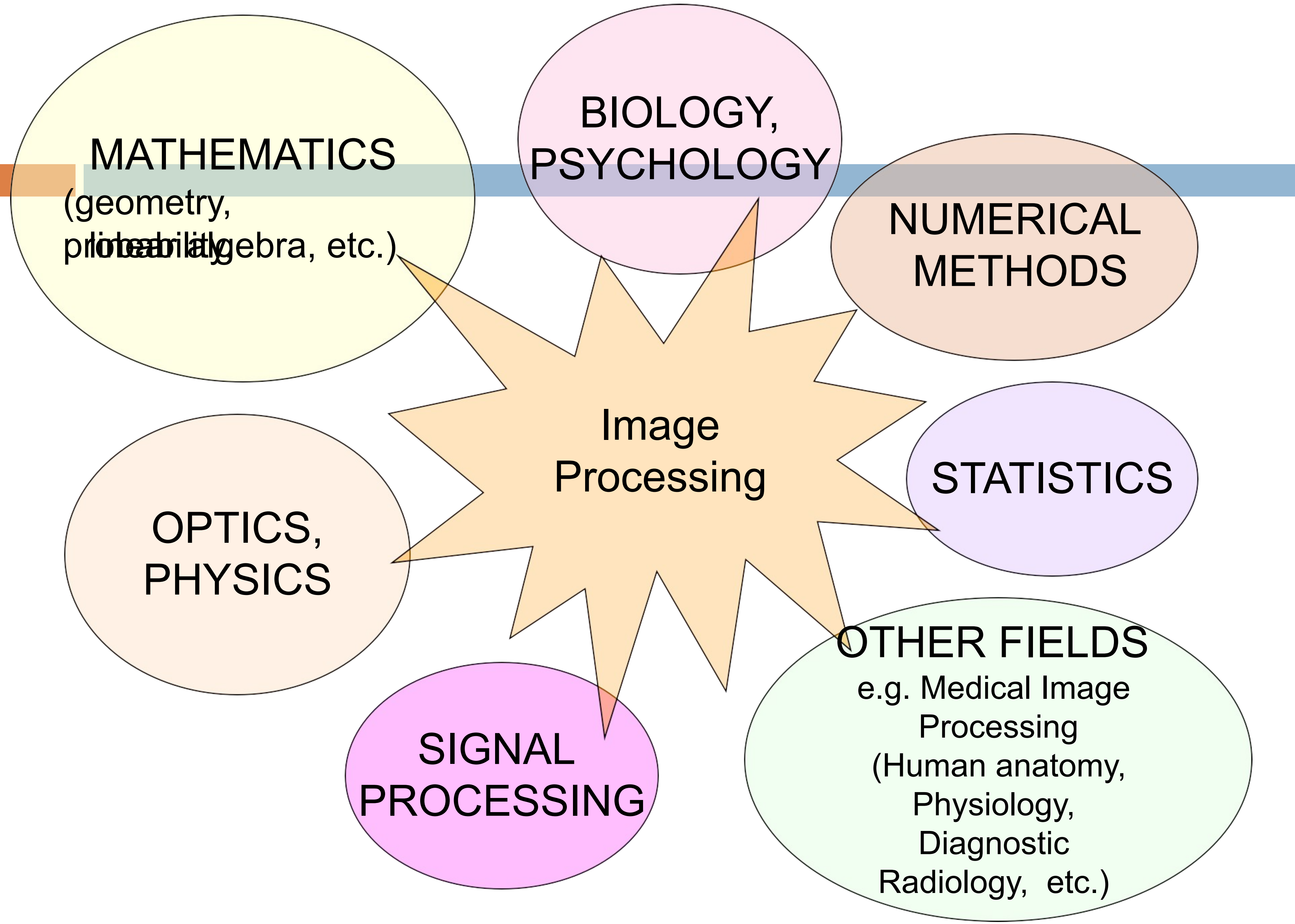
- **More specifically, DIP is**

- Improvement of pictorial forhuman information interpretation.
- Processing of image data for storage, transmission, and representation for autonomous machine perception.
- **No clear boundary as to where image processing ends and fields such as image analysis and computer vision start**

Image Processing Vs Computer Vision

The continuum from image processing to computer vision can be broken up into low-, mid- and high-level processes

Low Level Process	Mid Level Process	High Level Process
Input: Image **Output:** Image	**Input:** Image **Output:** Attributes	**Input:** Attributes **Output:** Understanding
Examples: Noise removal, image sharpening	**Examples:** Object recognition, segmentation	**Examples:** Scene understanding, autonomous navigation


MATHEMATICS
(geometry,
probability
linear algebra, etc.)
BIOLOGY, PSYCHOLOGY
NUMERICAL METHODS
Image Processing
STATISTICS
OPTICS, PHYSICS
SIGNAL PROCESSING
OTHER FIELDS
e.g. Medical Image Processing
(Human anatomy, Physiology, Diagnostic Radiology, etc.)

Quantization

Quantization is **a lossy compression technique which is achieved by compressing a range of values to single quantum**. In other words, we can also say that it is a process of converting a continuous range of values into a finite range of discrete values.

Sampling

The sampling rate determines the spatial resolution of the digitized image, while the quantization level determines the number of grey levels in the digitized image. A magnitude of the sampled image is expressed as a digital value in image processing. The transition between continuous values of the image function and its digital equivalent is called quantization.

History of Digital Image Processing

First Digital Photograph

In 1957 when Russell Kirsch made a 176×176 pixel **digital image**

History of Digital Image Processing

Early 1920s: One of the first applications of digital imaging was in the news- paper industry

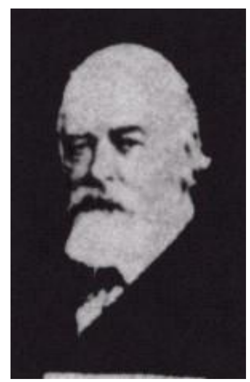
- The Bartlane cable picture transmission service
- Images were transferred by submarine cable between London and New York
- Pictures were coded for cable transfer and reconstructed at the receiving end on a telegraph printer

Early digital image

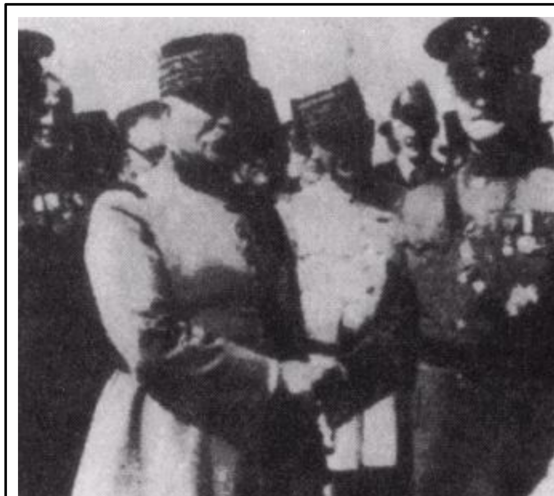
History of Digital Image Processing …

Mid to late 1920s: Improvements to the Bartlane system resulted in higher quality images

- New reproduction processes based on photographic techniques
- Increased number of tones in reproduced images



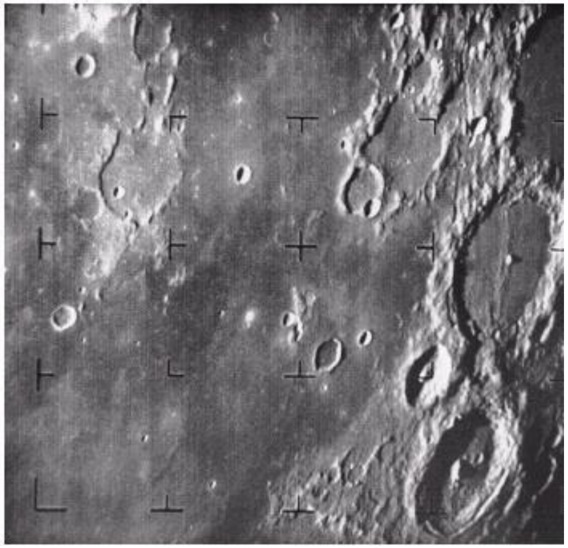
Improved digital image



History of Digital Image Processing …

1960s: Improvements in computing technology and the onset of the space race led to a surge of work in digital image processing

- **1964:** Computers used to improve the quality of images of the moon taken by the *Ranger 7* probe
- Such techniques were used in other space missions including the Apollo landings

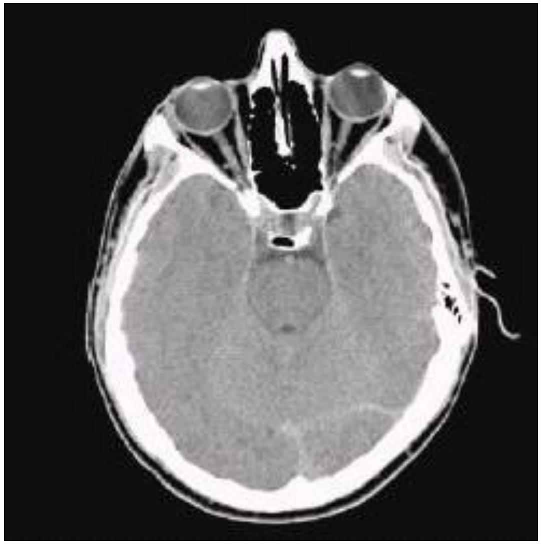


A picture of the moon taken by the Ranger 7 probe minutes before landing

History of Digital Image Processing …

1970s: Digital image processing begins to be used in medical applications

- **1979:** Sir Godfrey N. Hounsfield & Prof. Allan M. Cormack share the Nobel Prize in medicine for the invention of tomography, the technology behind Computerised Axial Tomography (CAT) scans



Typical head slice CAT image

History of Digital Image Processing …

1980s - Today: The use of digital image processing techniques has exploded and they are now used for all kinds of tasks in all kinds of areas

- Image enhancement/restoration
- Artistic effects
- Medical visualisation
- Industrial inspection
- Law enforcement
- Human computer interfaces

Why do we need Image Processing?

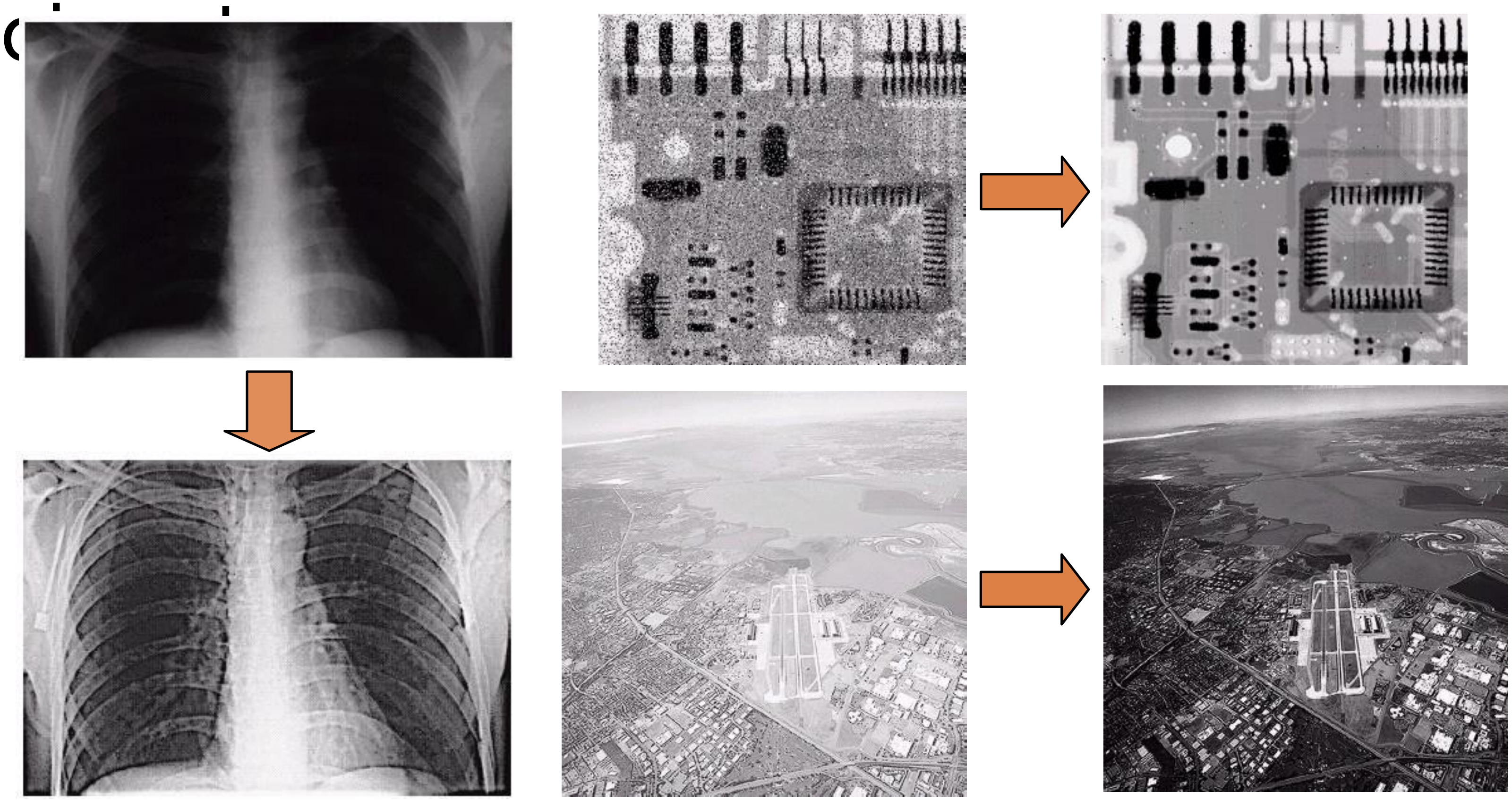
- What do we mean by *Digital Image Processing*
  - **Processing digital images by a digital computer**

It is Motivated by three major applications-

- Improvement of pictorial information for human perception
- Image processing for autonomous machine application
- Efficient storage and transmission

Examples: Image Enhancement

One of the most common uses of DIP techniques: improve quality, remove no



Human Perception

Employ methods capable of **enhancing pictorial information** for human interpretation and analysis

- Typical applications:
  - Noise filtering
  - Content enhancement
    - Contrast enhancement
    - Deblurring
  - Remote sensing

Filterin
g

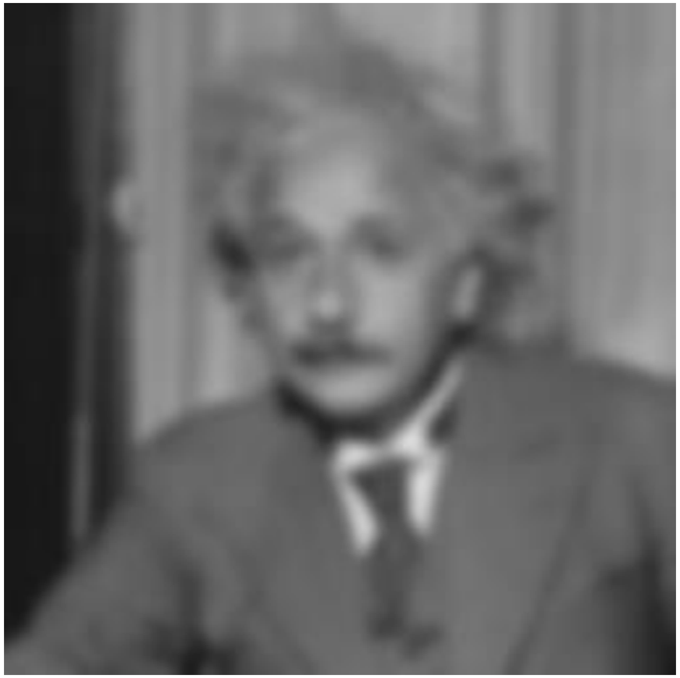
Filterin
g

Noisy Image

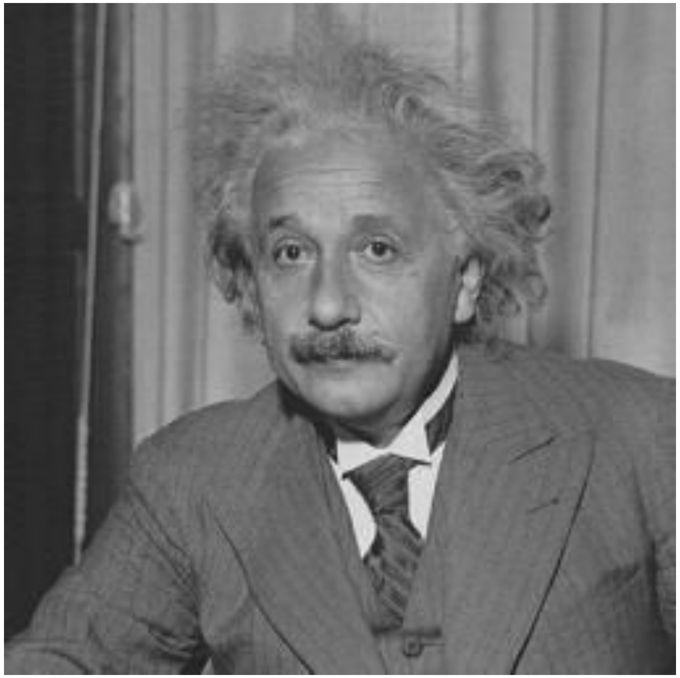
Filtered Image

Image Enhancement

Image Deblurring



Deblurring



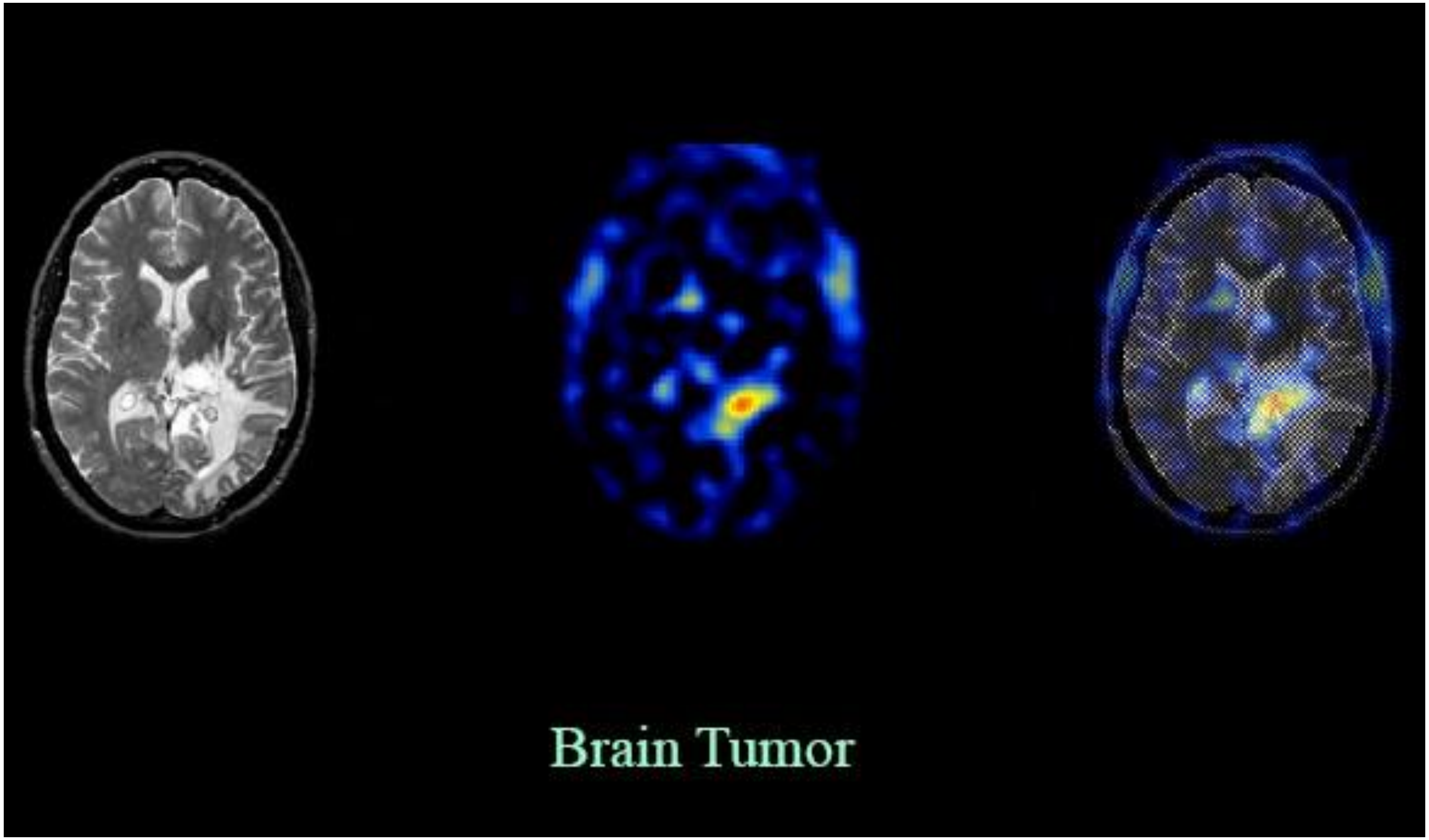
Blurred

Deblurred

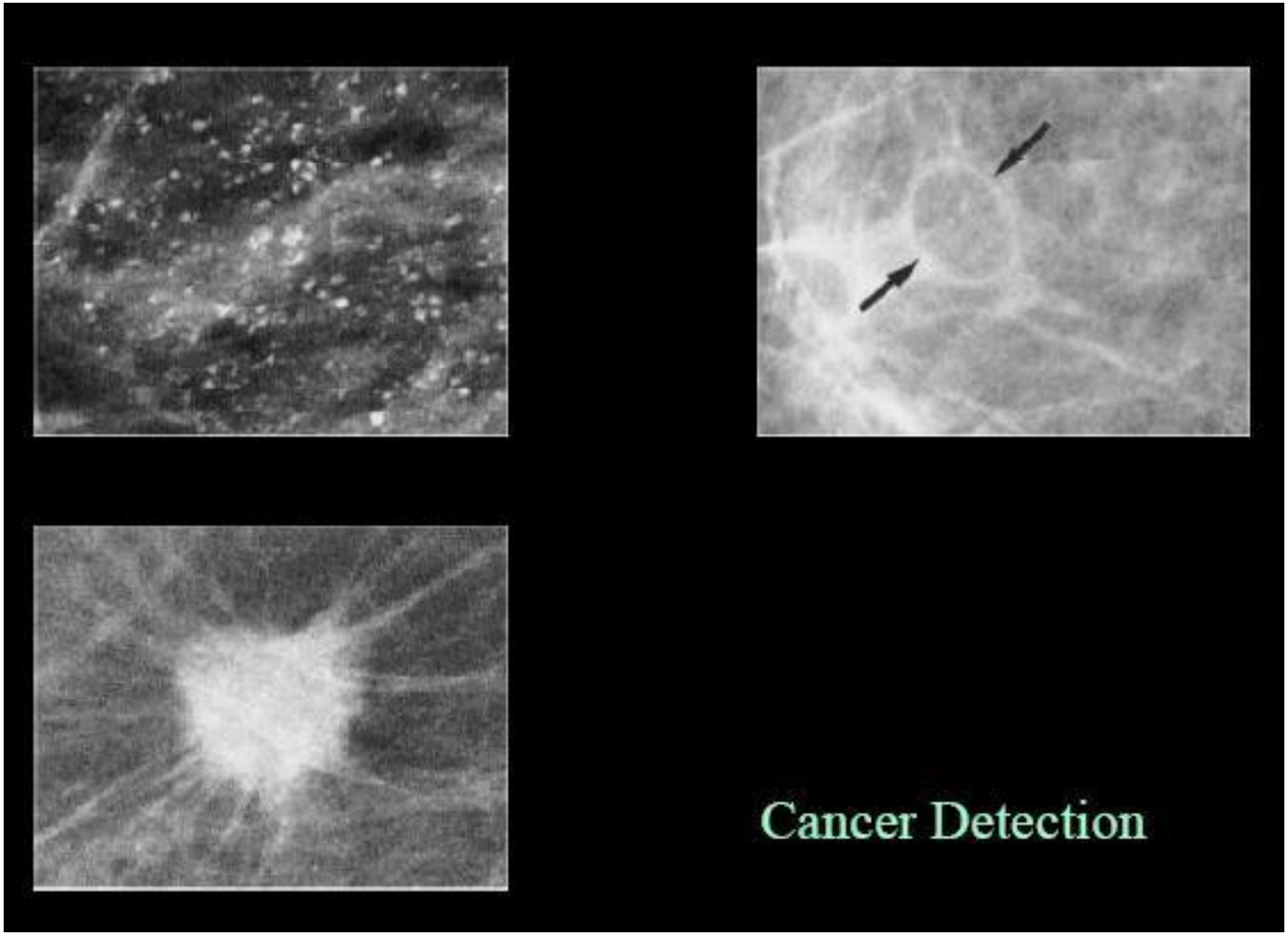
The Hubble Telescope

- Launched in 1990 the Hubble telescope can take images of very distant objects
- However, an incorrect mirror made many of Hubble's images usele
- Image processing techniques were used to fix this

Medical Imaging

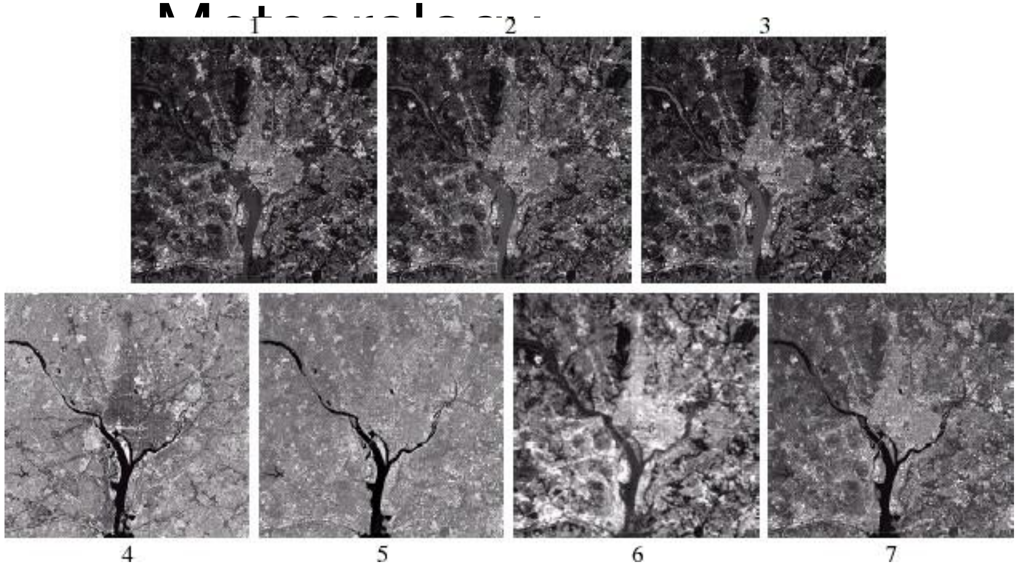


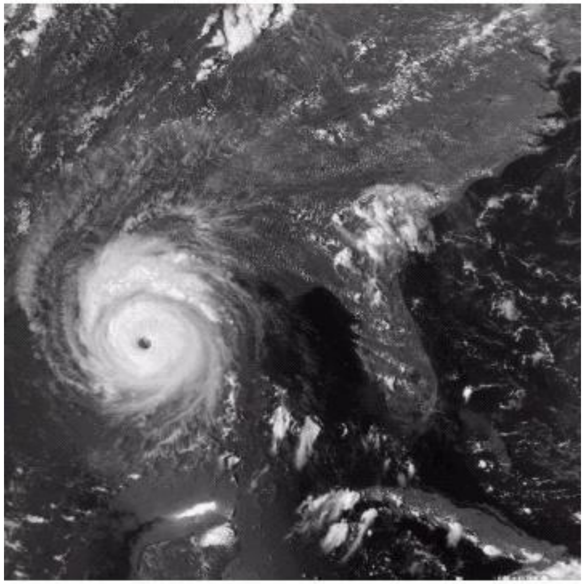
Medical Imaging



GIS

- Geographic Information Systems
  - Digital image processing techniques are used extensively to manipulate satellite imagery
  - Terrain classification
  - Meteorology

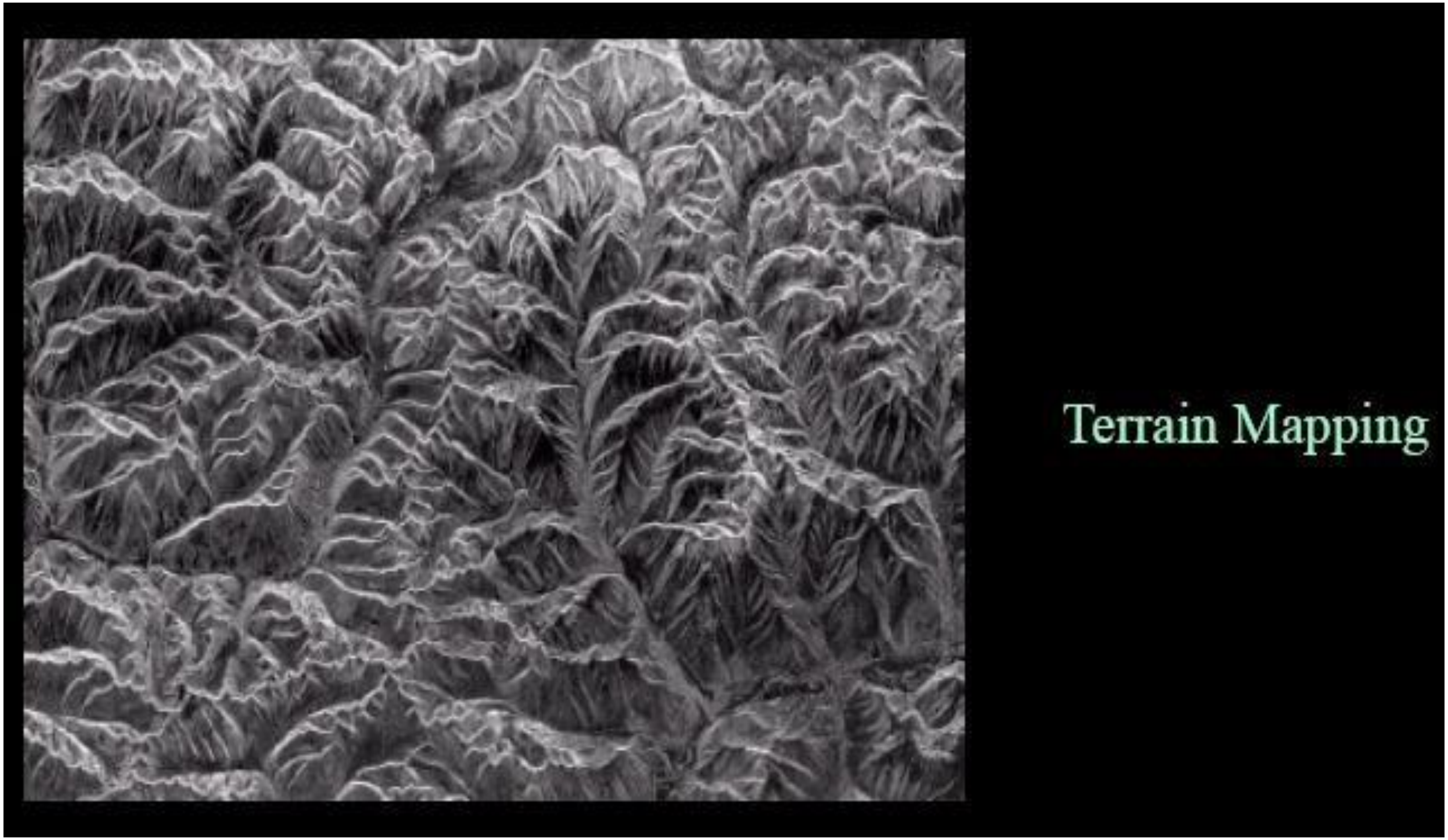




Remote Sensing

Satellite Image, Kolkata

Remote Sensing …



Weather Forecasting

Astronomy

Artistic Effects

- Artistic effects are used to make images more visually appealing, to add special effects and to make composite images

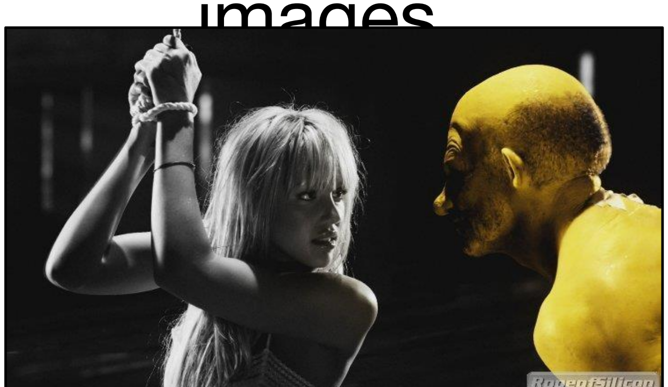


Image Compression

- An image usually contains lot of redundancy that can be exploited to achieve compression
  - Pixel redundancy
  - Coding redundancy
  - Psychovisual redundancy
- Applications:
  - Reduced storage
  - Reduction in bandwidth

Image Compression …

Full quality (Q = 100), filesize 83261 B.   High quality (Q = 50), filesize 15138 B.

Image Compression …

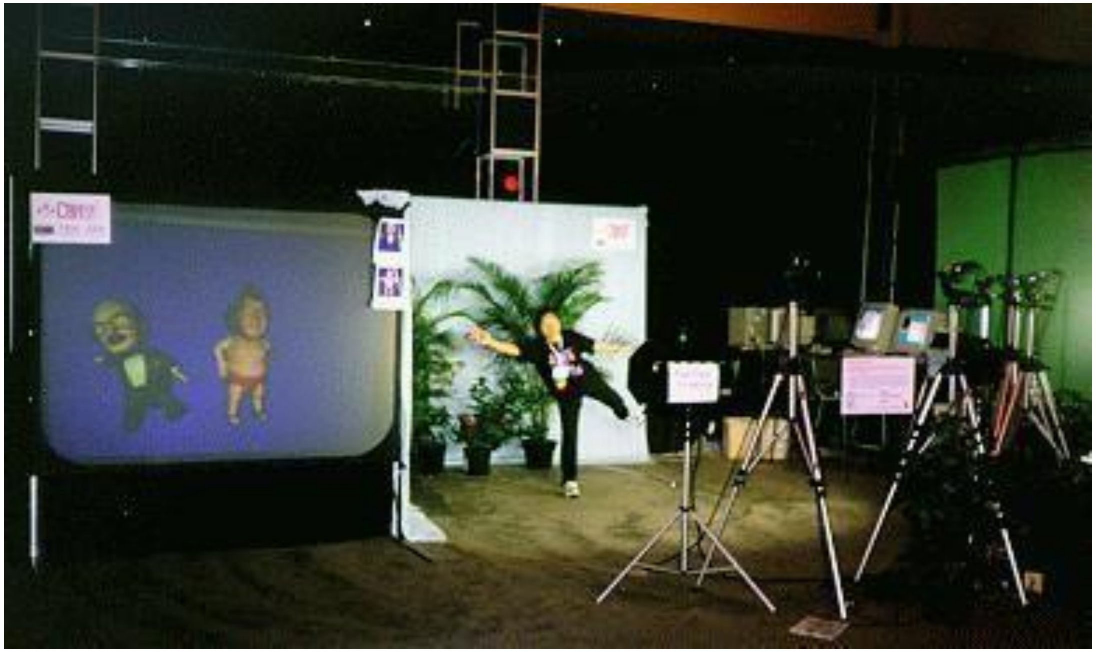
Medium quality (Q = 25), filesize 9553 B. Low quality (Q = 10), filesize 4787 B.

Image Compression ...

Low quality (Q = 10), filesize 4787 B. Lowest quality (Q = 1), filesize 1523 B.

What is Computer Vision?

- Deals with the development of the **theoretical** and **algorithmic** basis by which useful information about the 3D world can be automatically extracted and analyzed from a **single** or **multiple** 2D images of the world.



Why is Computer Vision Difficult?

- It is a **many-to-one** mapping
  - A variety of surfaces with different *material* and *geometrical* properties, possibly under different *lighting* conditions, could lead to identical images
  - Inverse mapping has non unique solution (a lot of information is **lost** in the transformation from the 3D world to the 2D image)
- It is **computationally intensive**
- We **do not understand** the recognition problem

Recognition Cues

Scene interpretation, even of complex, cluttered scenes is a straightforward task for humans

Recognition Cues (cont'd)

How are we able to discern reality and an image of reality?

What clues are present in the image?

What ... age?

The role of color

What is this object?
Does color play a role in recognition?
Might this be easier to recognize from a different view?

The role of texture

- Characteristic image texture can help us readily
  recognize objects.

The role of shape

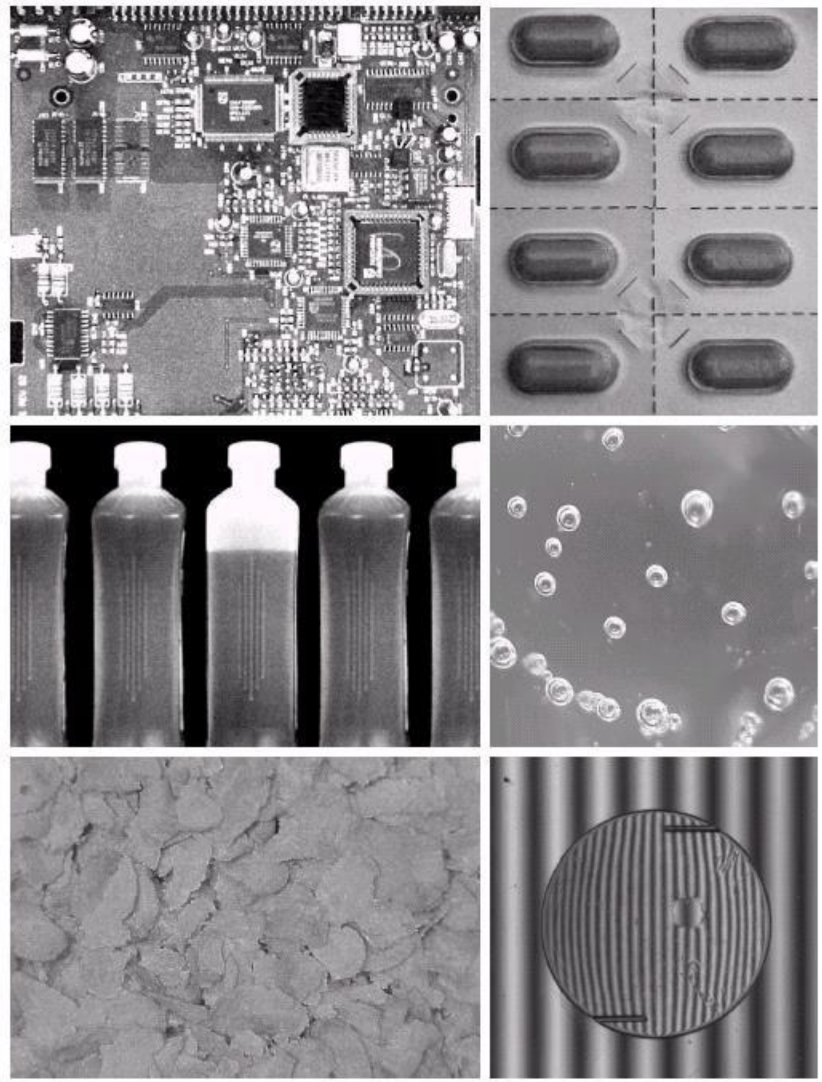
The role of grouping

Computer Vision Applications

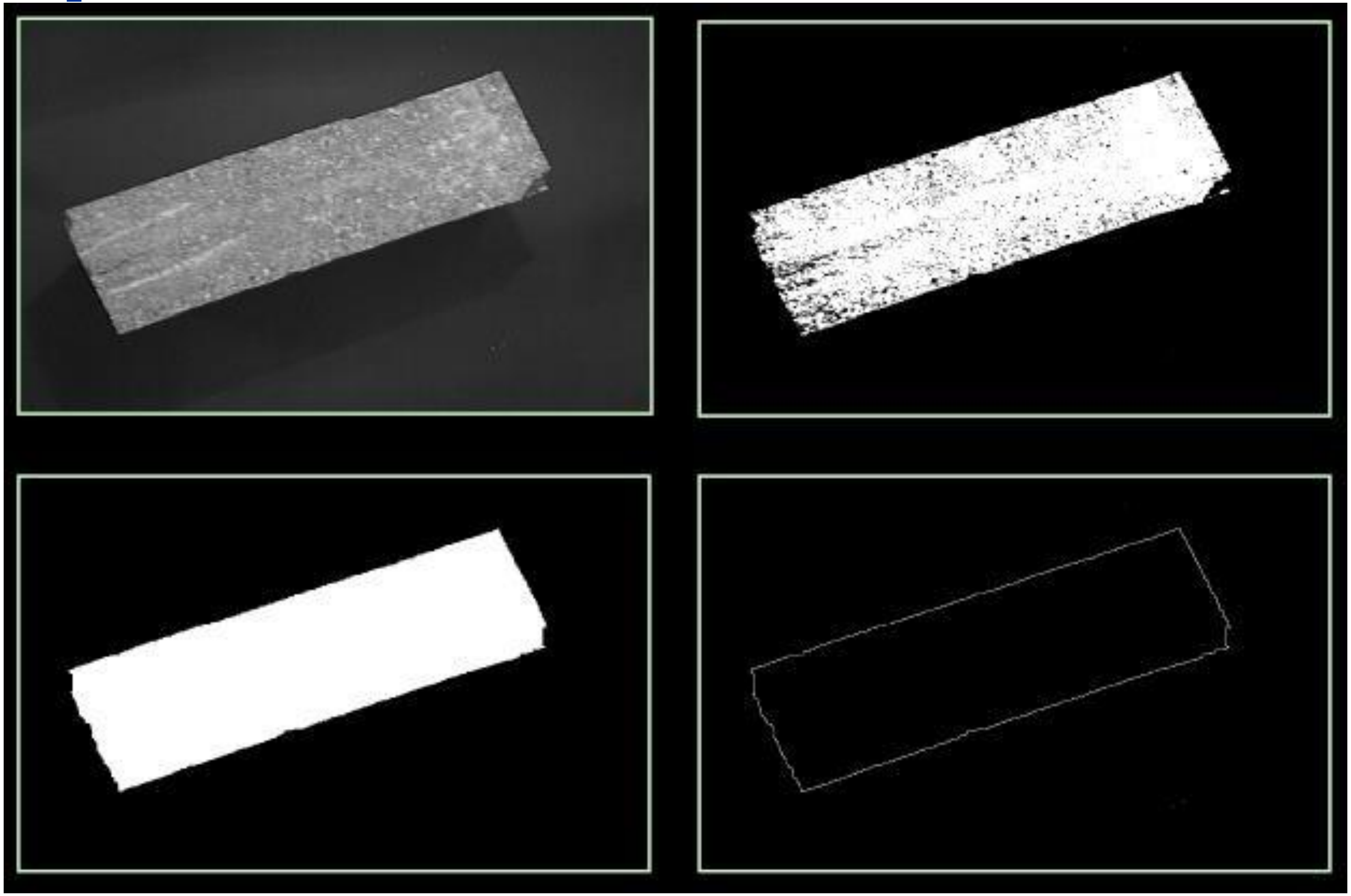
- Industrial inspection/quality control
- Surveillance and security
- Face recognition
- Gesture recognition
- Space applications
- Medical image analysis
- Autonomous vehicles
- Virtual reality and much more ……

Examples: Industrial Inspection

- Human operators are expensive, slow and unreliable
- Make machines do the job instead
- Industrial vision systems are used in all kinds of industries
  - Can we trust them?

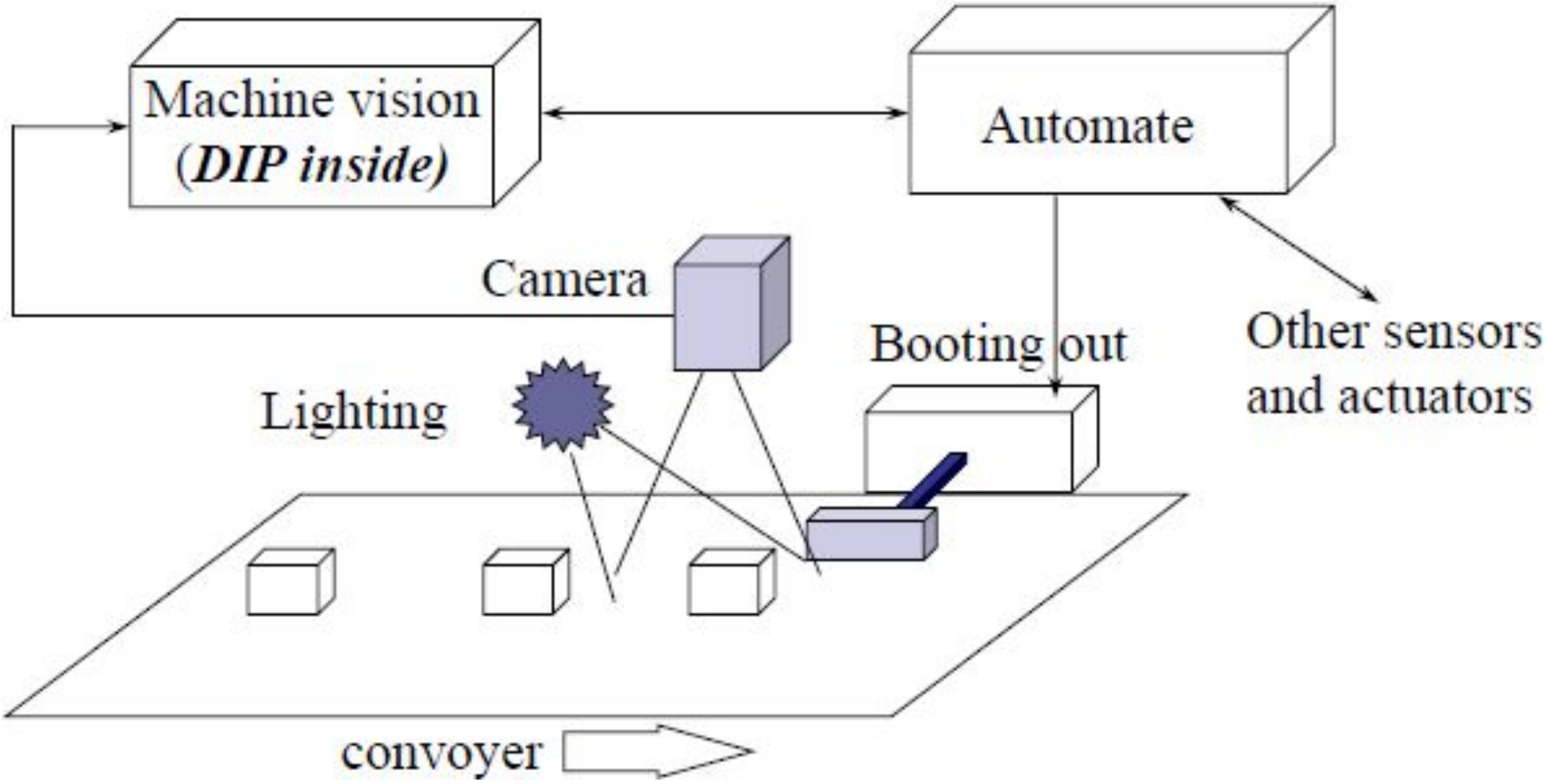


Automated Inspection …



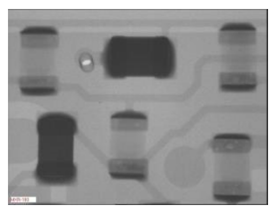
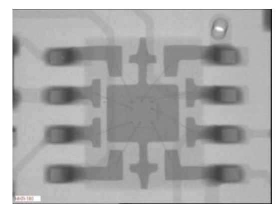
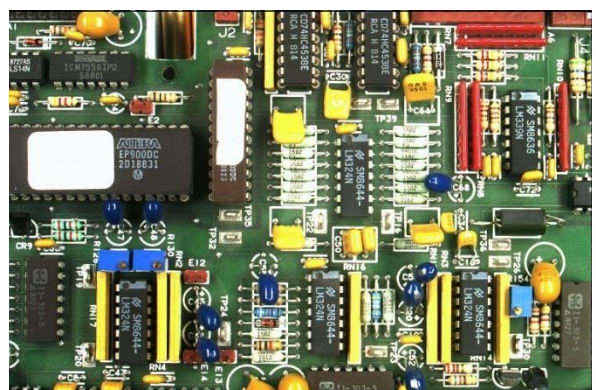
Automated Inspection …

- Industrial inspection, computer vision

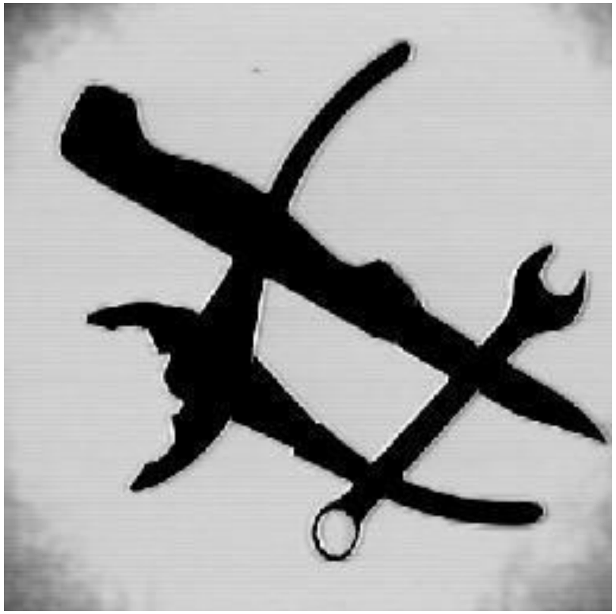


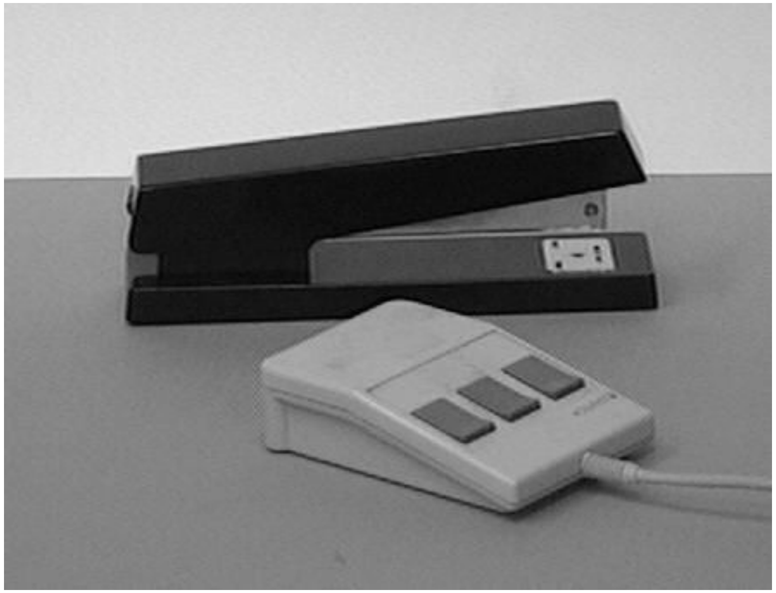
Examples: PCB Inspection

- Printed Circuit Board (PCB) inspection
  - Machine inspection is used to determine that all components are present and that all solder joints are acceptable
  - Both conventional imaging and x-ray imaging are

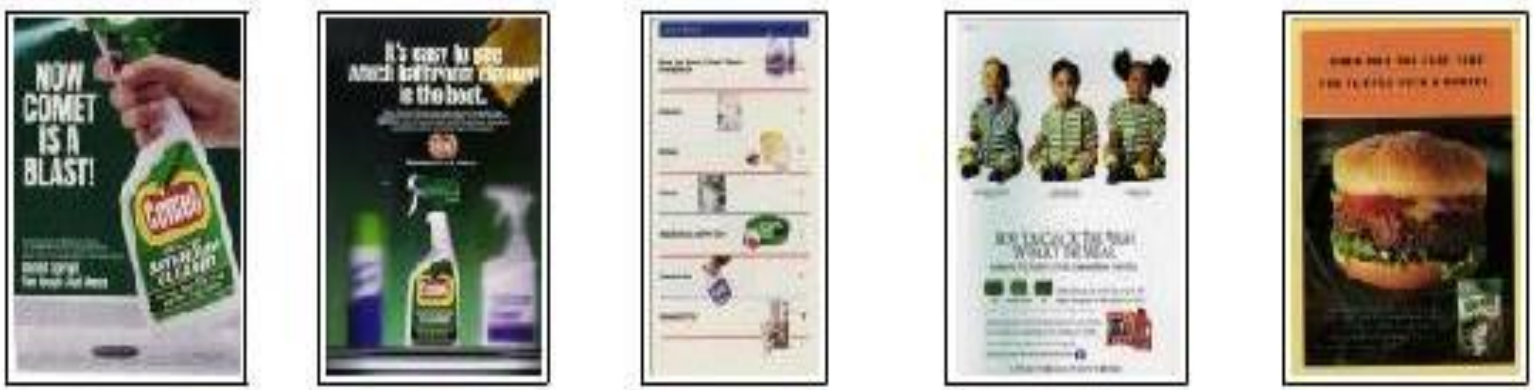

Object Recognition





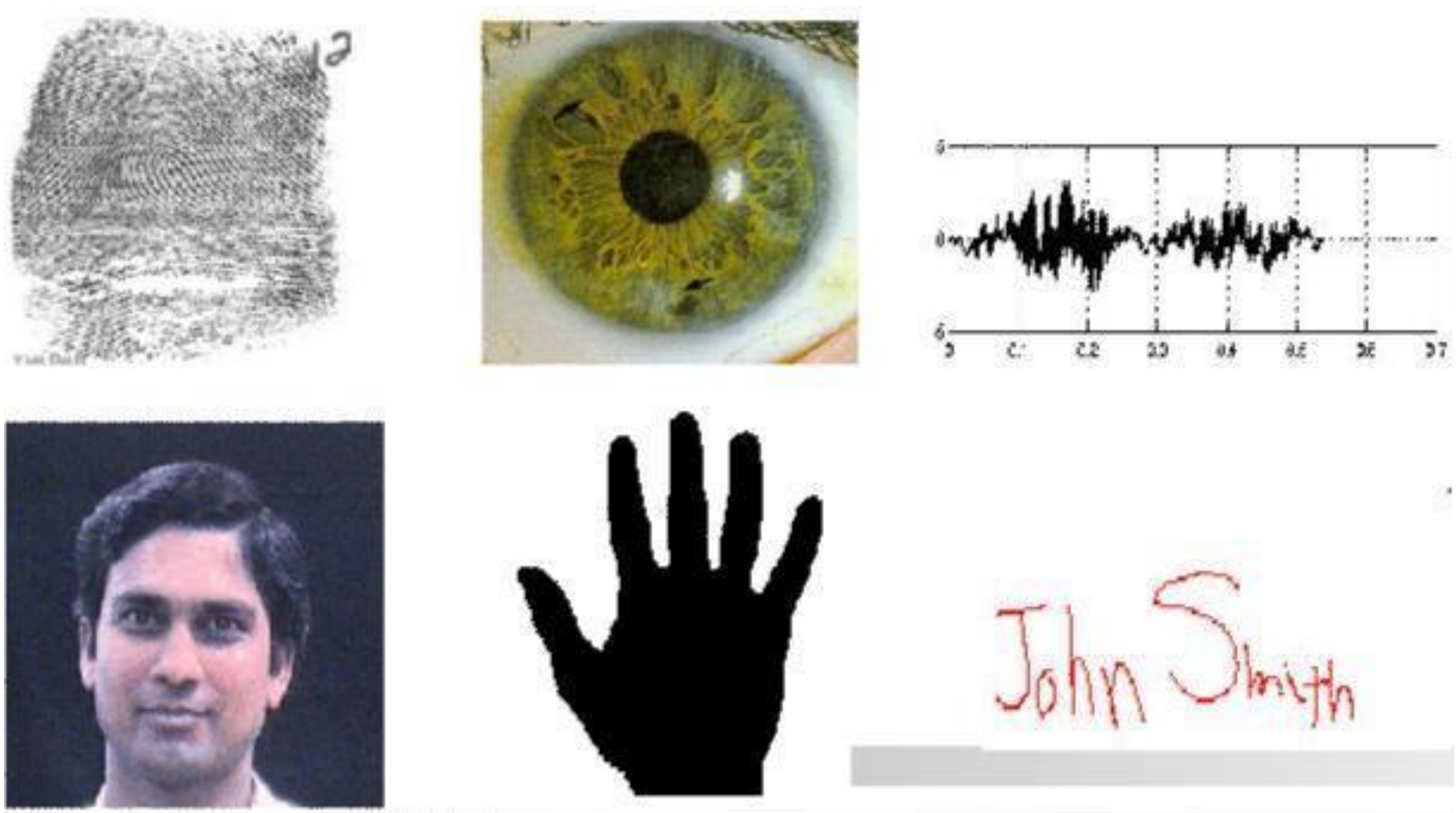
Indexing into Databases (cont'd)

- Color, texture

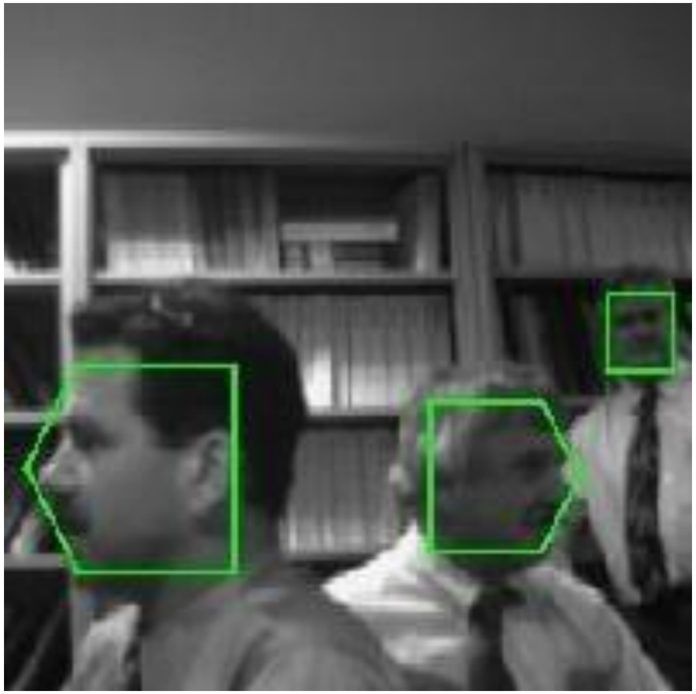


$T = 33.6\text{s}$, found 2 of 2

Biometrics



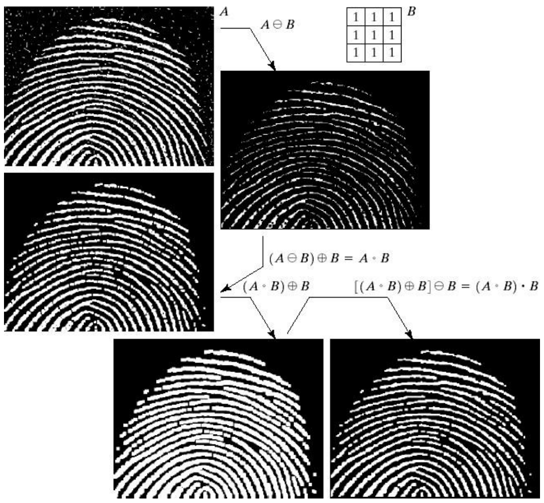
Face Detection



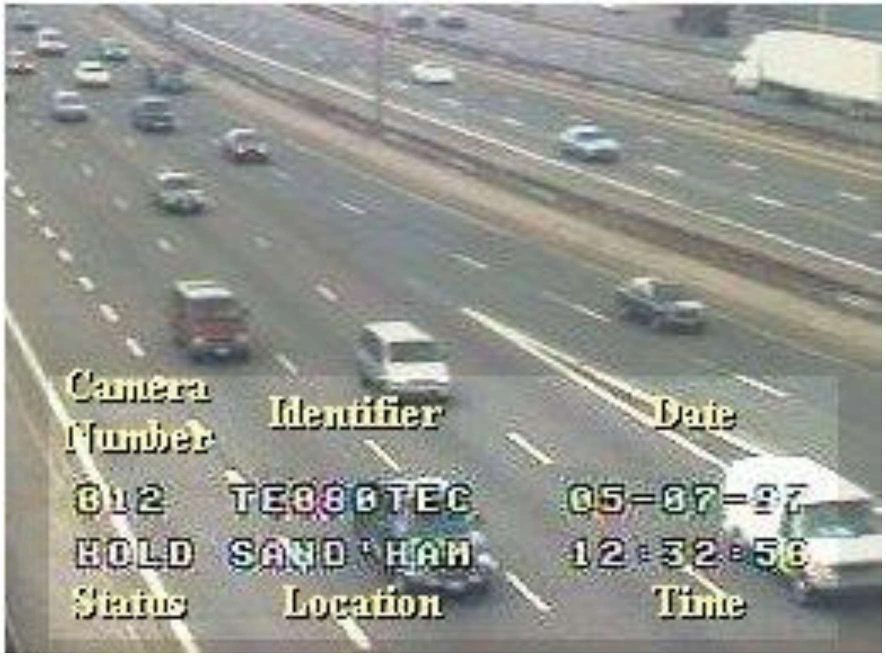
Face Recognition

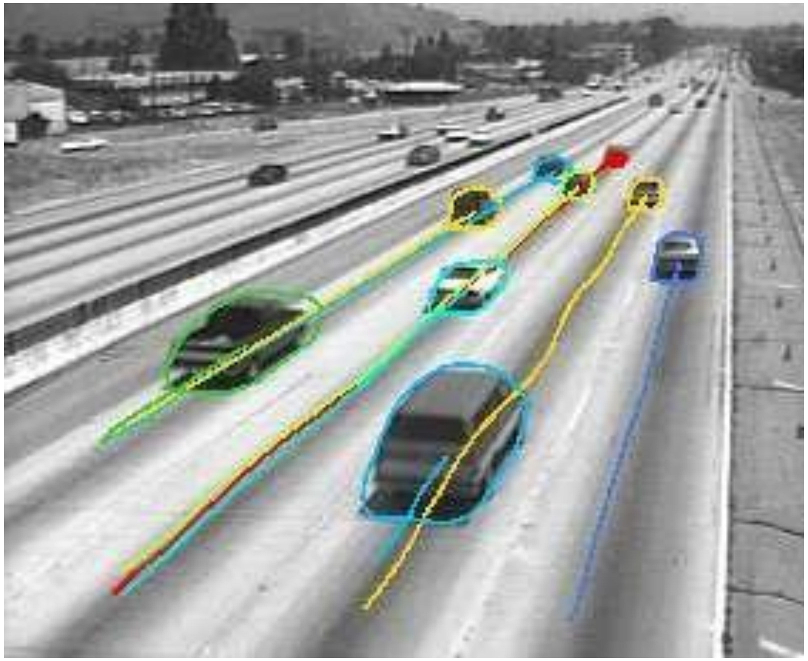
Examples: Law Enforcement

- Image processing techniques are used extensively by law enforcers
  - Number plate recognition for speed cameras/automated toll systems
  - Fingerprint recognition
  - Enhancement of CCTV images



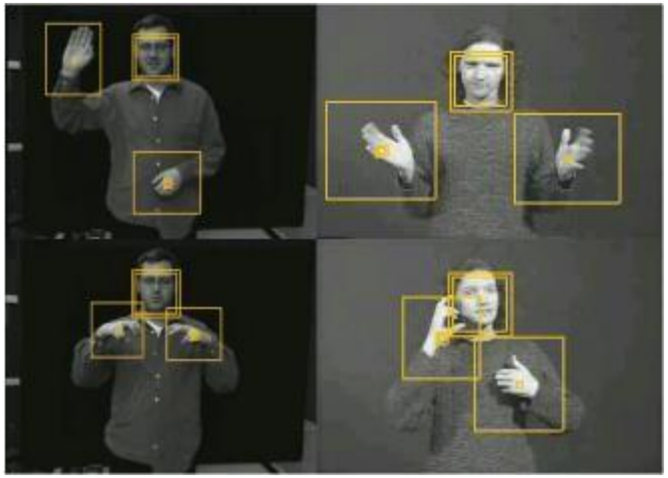
Traffic Monitoring

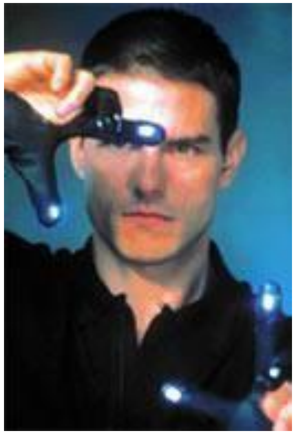


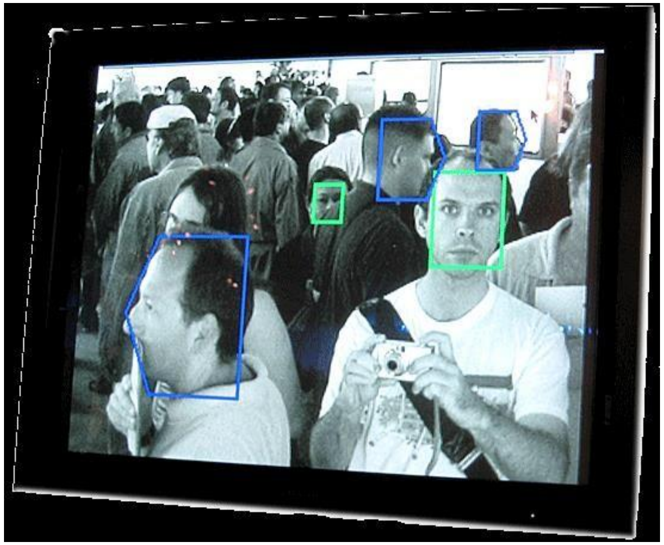


Examples: HCI

- Try to make human computer interfaces more natural
  - Face recognition
  - Gesture recognition
- These tasks can be difficult







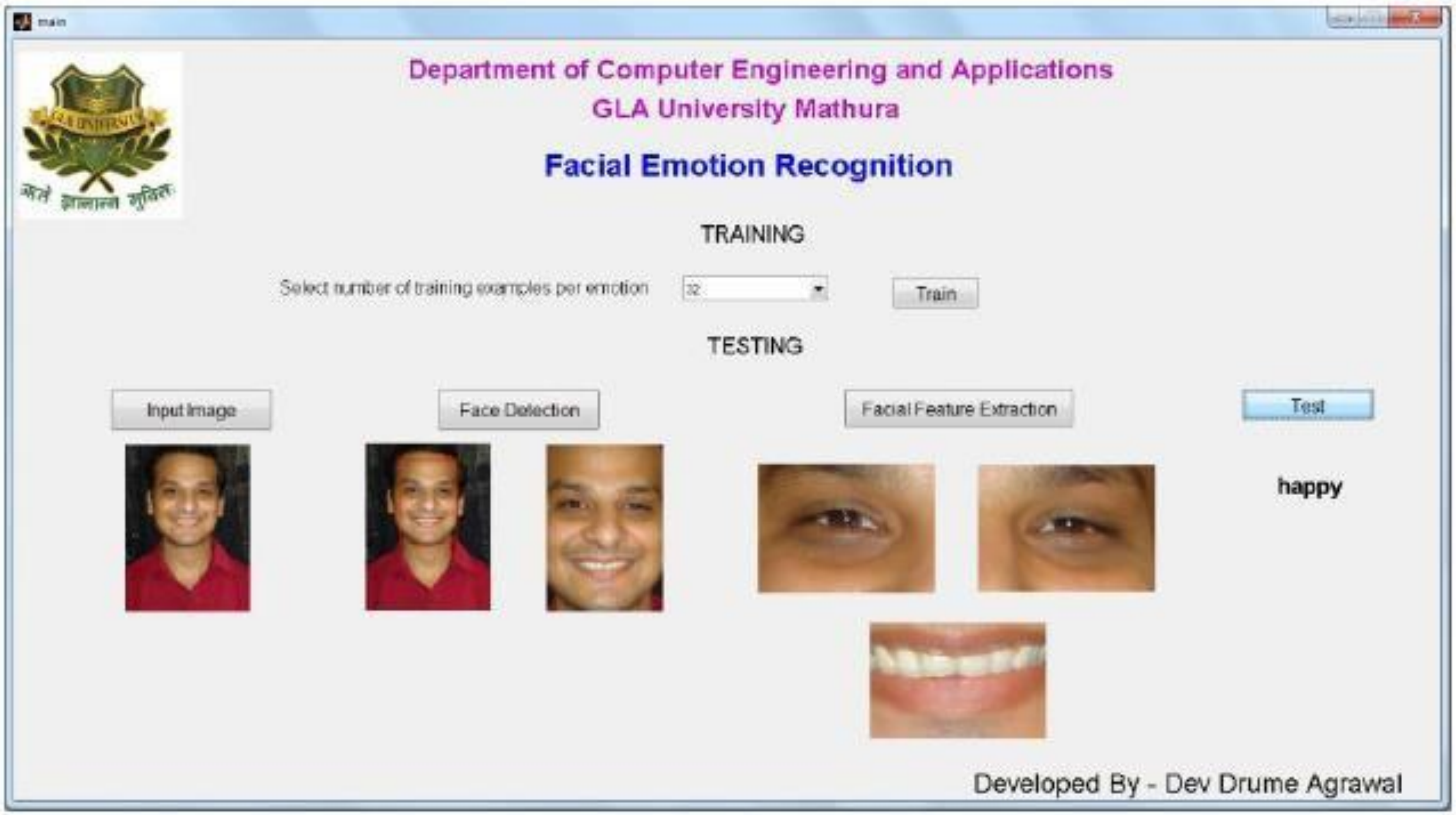
Computer Vision Applications …

- Video surveillance is the task of analyzing video data to identify suspicious activities in security-sensitive areas such as banks, department stores, parking lots.

- Manual surveillance requires the system to be monitored continuously by a person and is costly and problematic.

Computer Vision @ GLA

Computer Vision @ GLA …

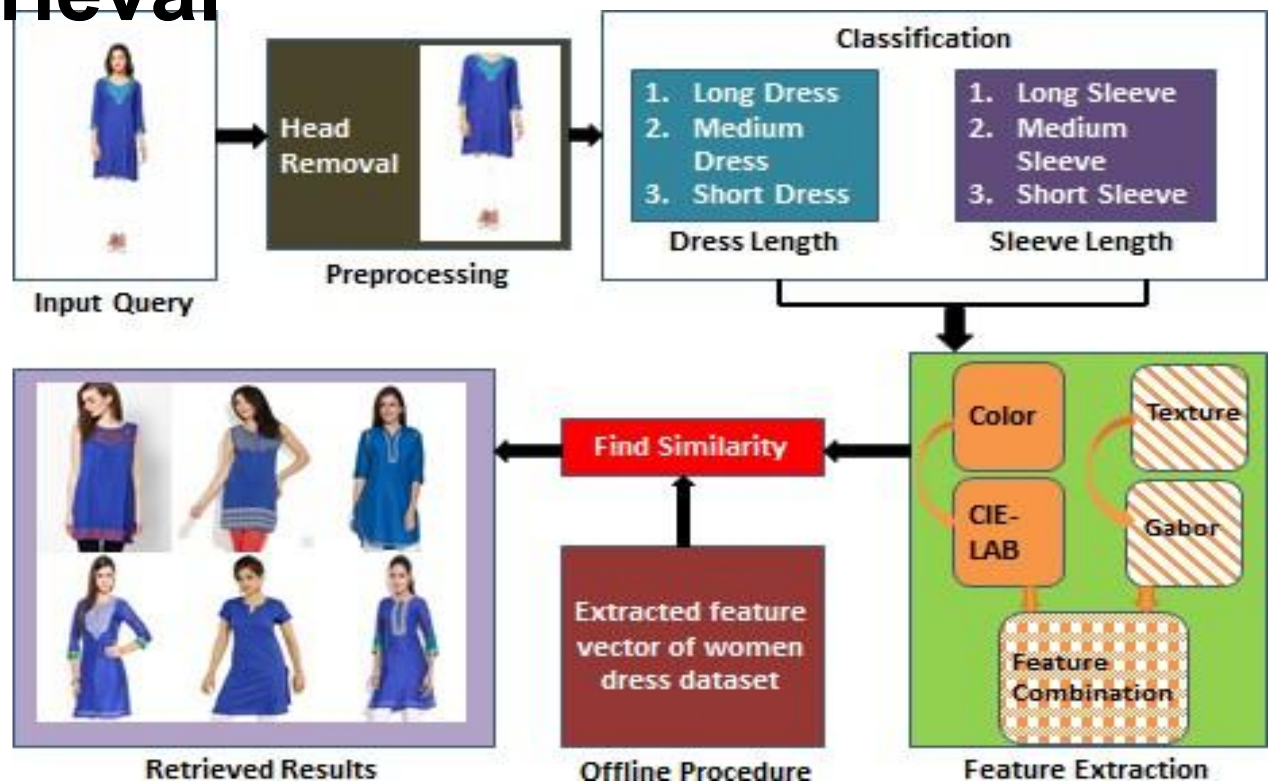


Computer Vision @ GLA …

Computer Vision @ GLA …

Computer Vision @ GLA …

Clothing Image Retrieval



Computer Vision @ GLA ...

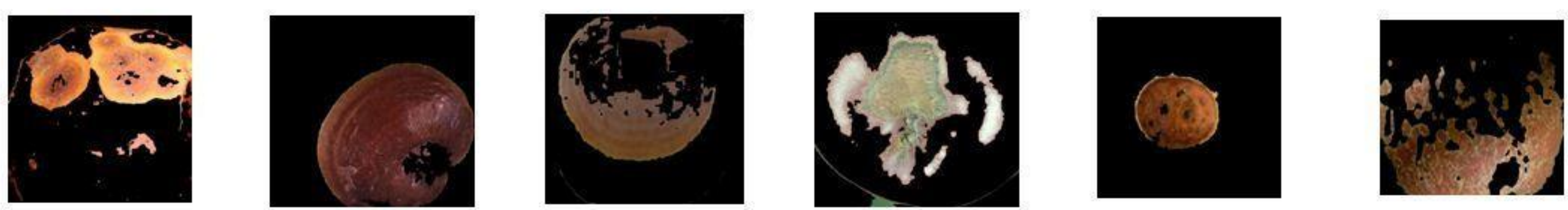
- **Applications in Agriculture**
  - Fruit and Vegetable Classification

Computer Vision @ GLA …

- **Applications in Agriculture …**

  Automatic Detection and Classification of Fruit Diseases

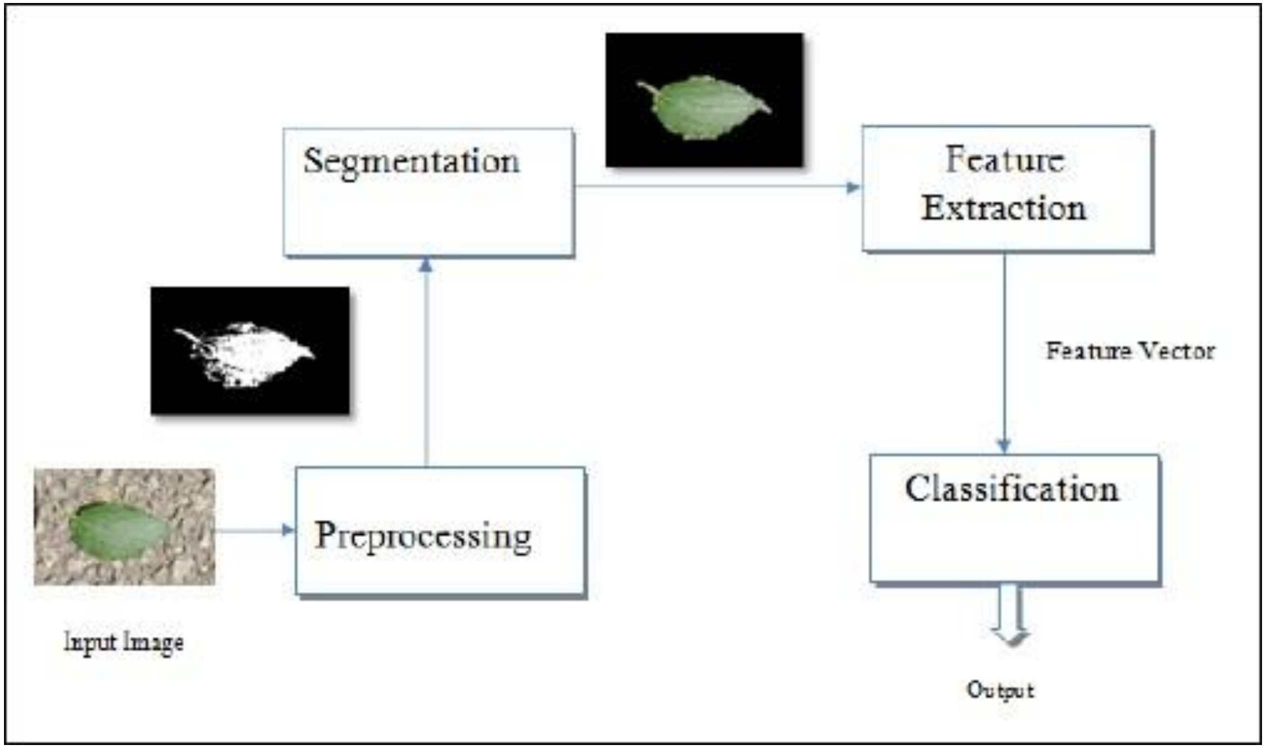
(a)



(b)

Computer Vision @ GLA …

Plant Identification system

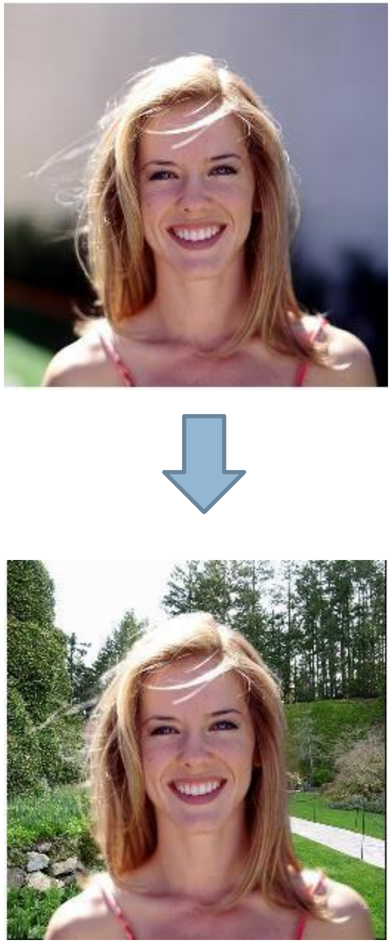


Computer Vision @ GLA ...

Salient Region Detection

Computer Vision @ GLA …

Digital image matting is a way through which we can extract foreground object from the given image.



Computer Vision @ GLA …

Image Forgery Detection

Computer Vision @ GLA …

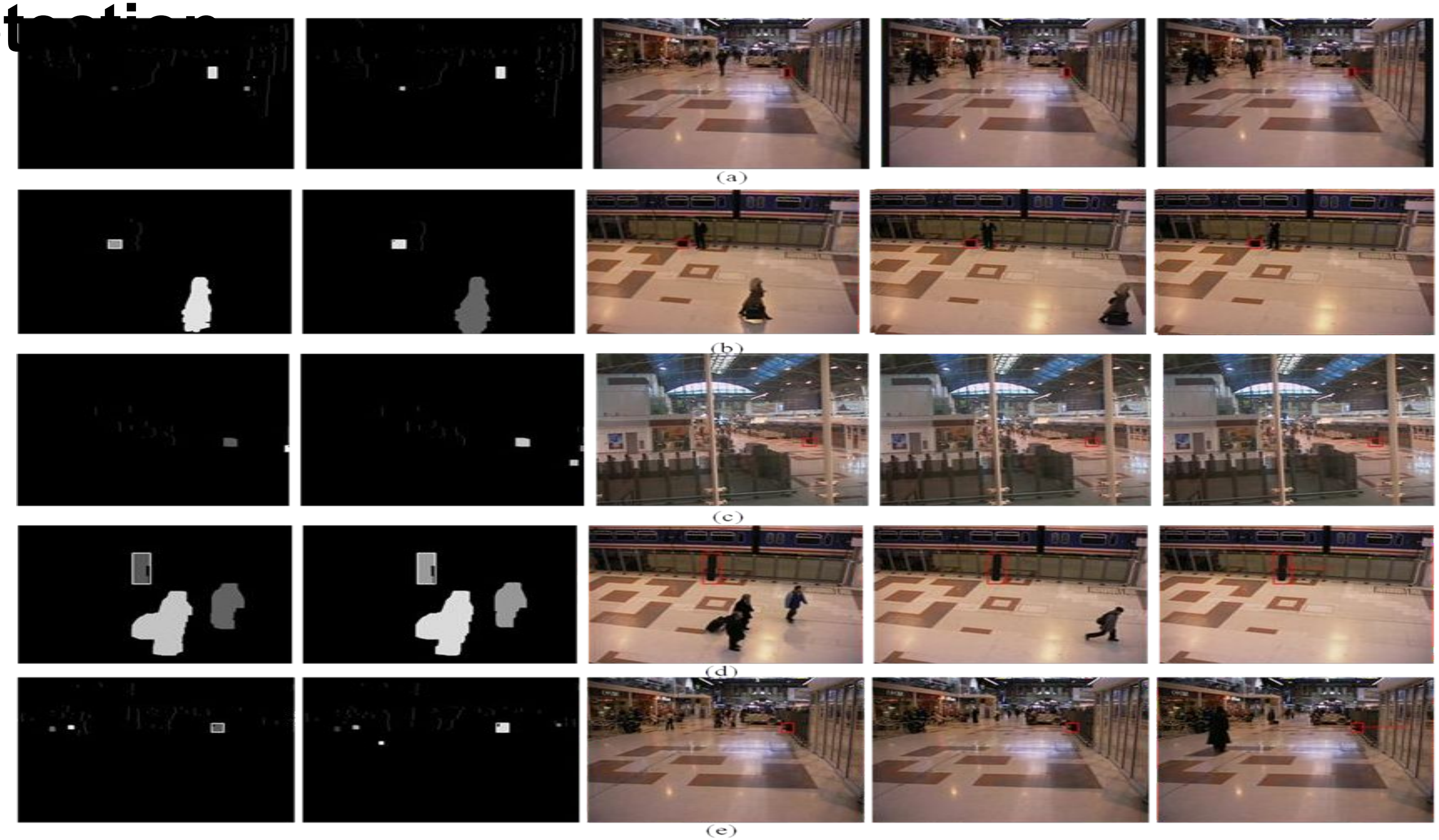
- Object Tracking

Computer Vision @ GLA ...

Human Activity Identification

Computer Vision @ GLA ...

Abandoned Object Detection



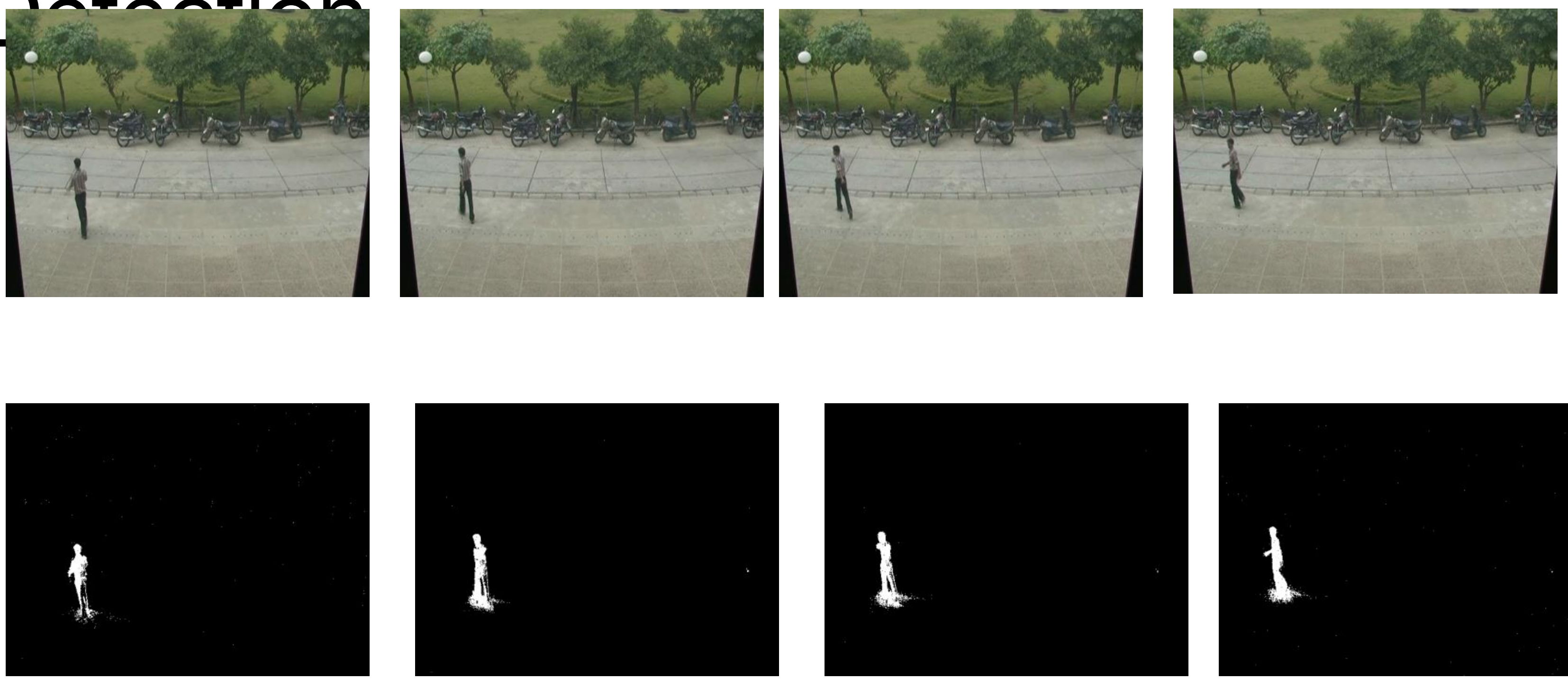
Machine Vision Applications …

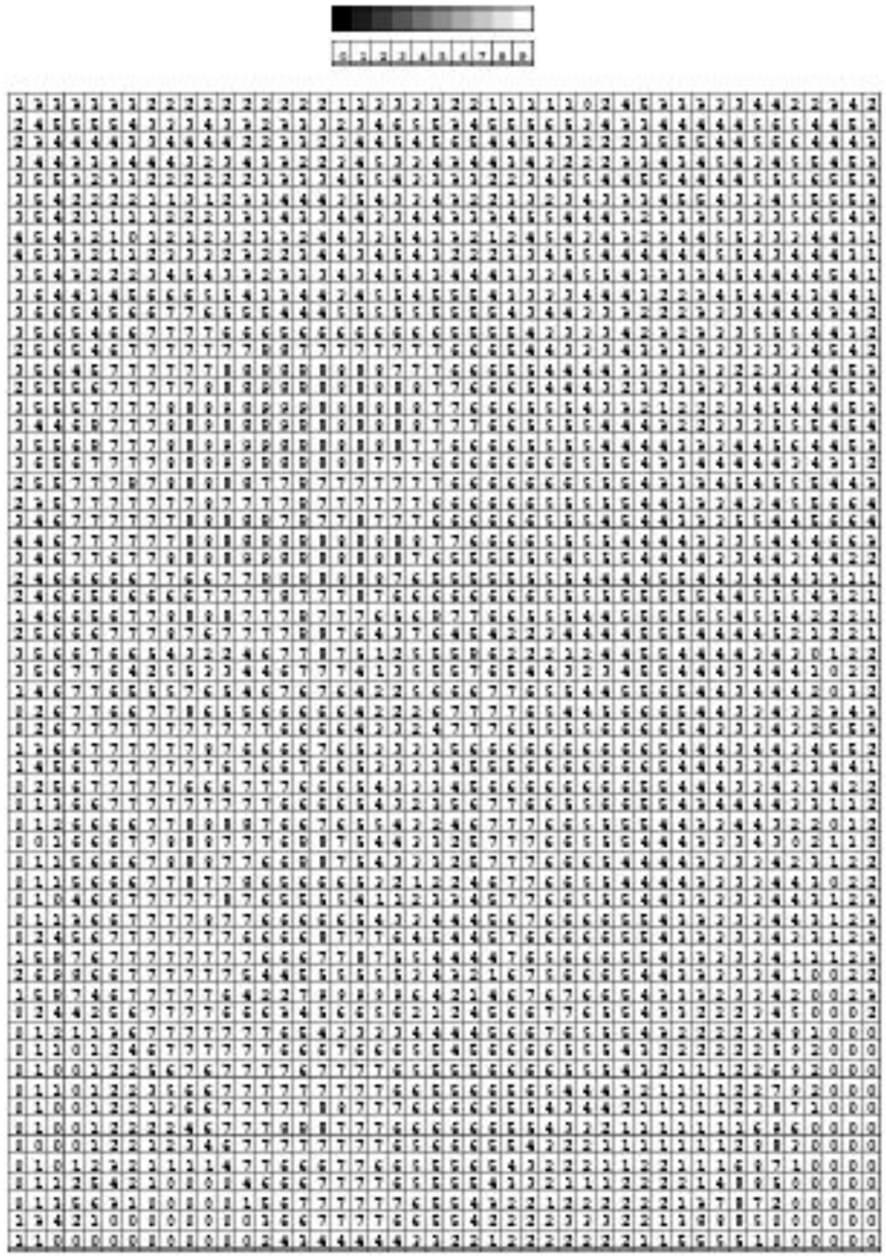
Video Sequence Processing

- The major emphasis of image sequence processing is detection of moving parts
- This has various applications
  - Detection and tracking of moving targets for security surveillance purpose
  - To find out the trajectory of a moving target
  - Monitoring the movements of organ boundaries in medical applications etc.

Machine Vision Applications …

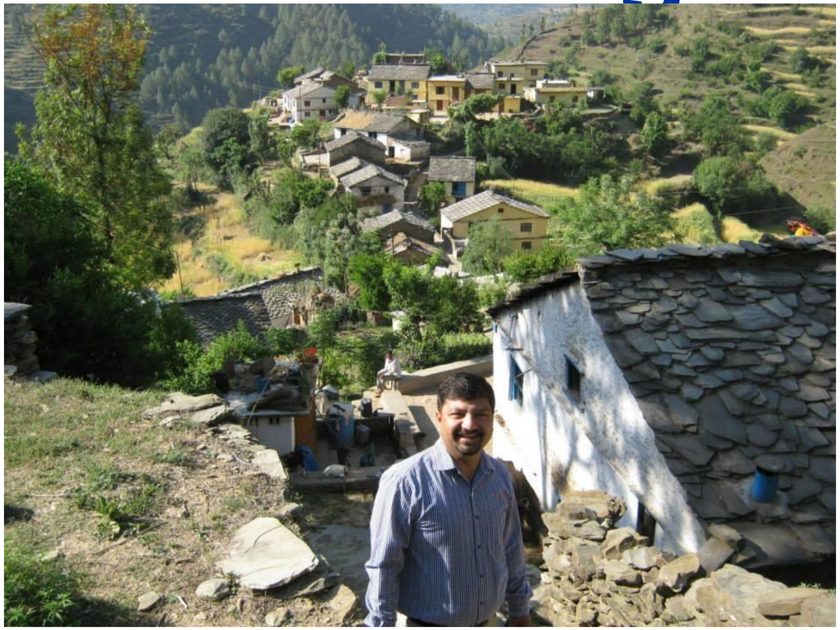
Movement Detection




0 1 2 3 4 5 6 7 8 9

Importance of prior information

Digital Image Processing



00000000	01110010	01100100	...
00011001	01000010	01100101	...
00011111	00110010	01100101	...
00000000	01110010	01100100	...
00011001	01000010	01100101	...
00011111	00110010	01100101	...
00000000	01110010	01100100	...
00011001	01000010	01100101	...
00011111	00110010	01100101	...
00000000	01110010	01100100	...
00011001	01000010	01100101	...
00011111	00110010	01100101	...
...			
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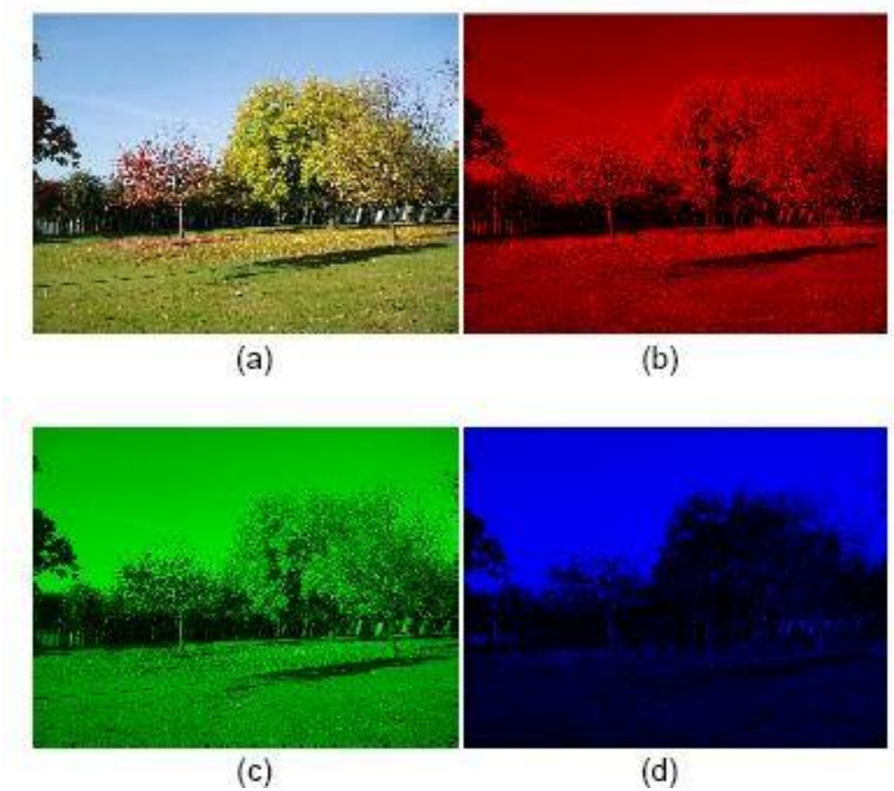
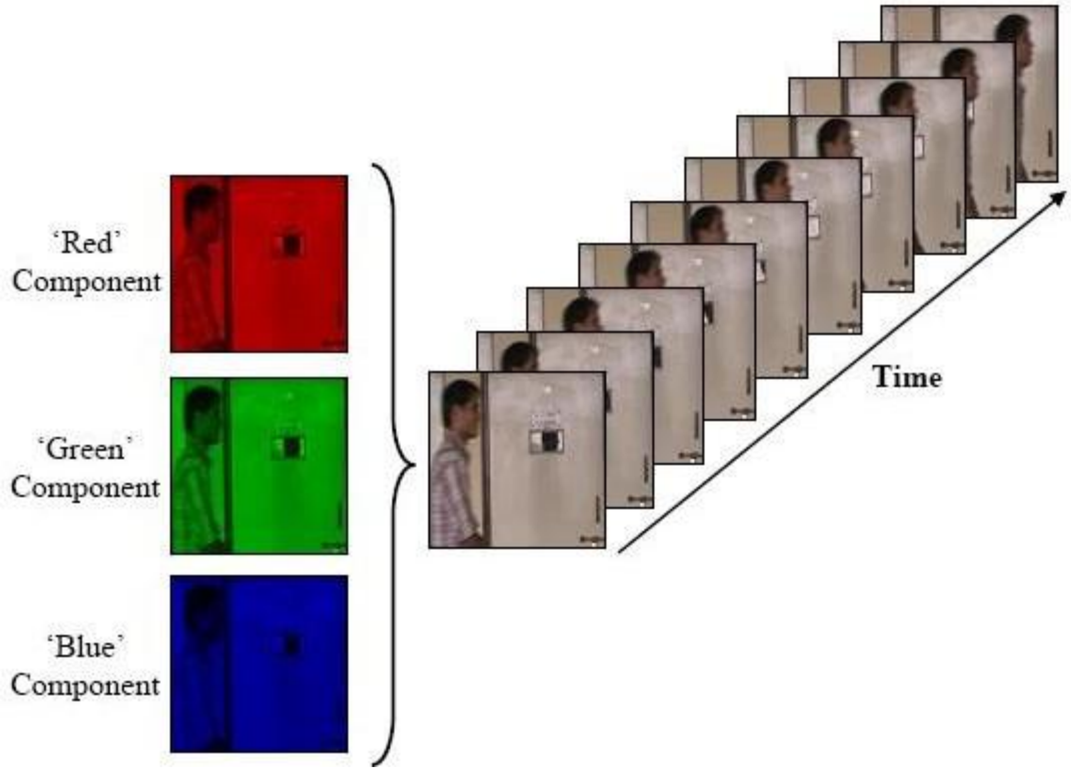
What we see

What the computer sees!!!

Digital image is a representation of a two- dimensional image using ones and zeros (binary).



RGB image along with its R, G, B components:
a) RGB
image b) R component c) G component d) B
component



Video structure and representation

Digital Image Processing …

Radiant (light) energy is recorded at corresponding points on a plane to

B/W images can be represented as a function of two variable,

$$f(x, y),$$

where f(x, y) is the brightness (or grayness or intensity)

In colour image, the intensity is measured in 3 wavelengths (red, green, blue), so

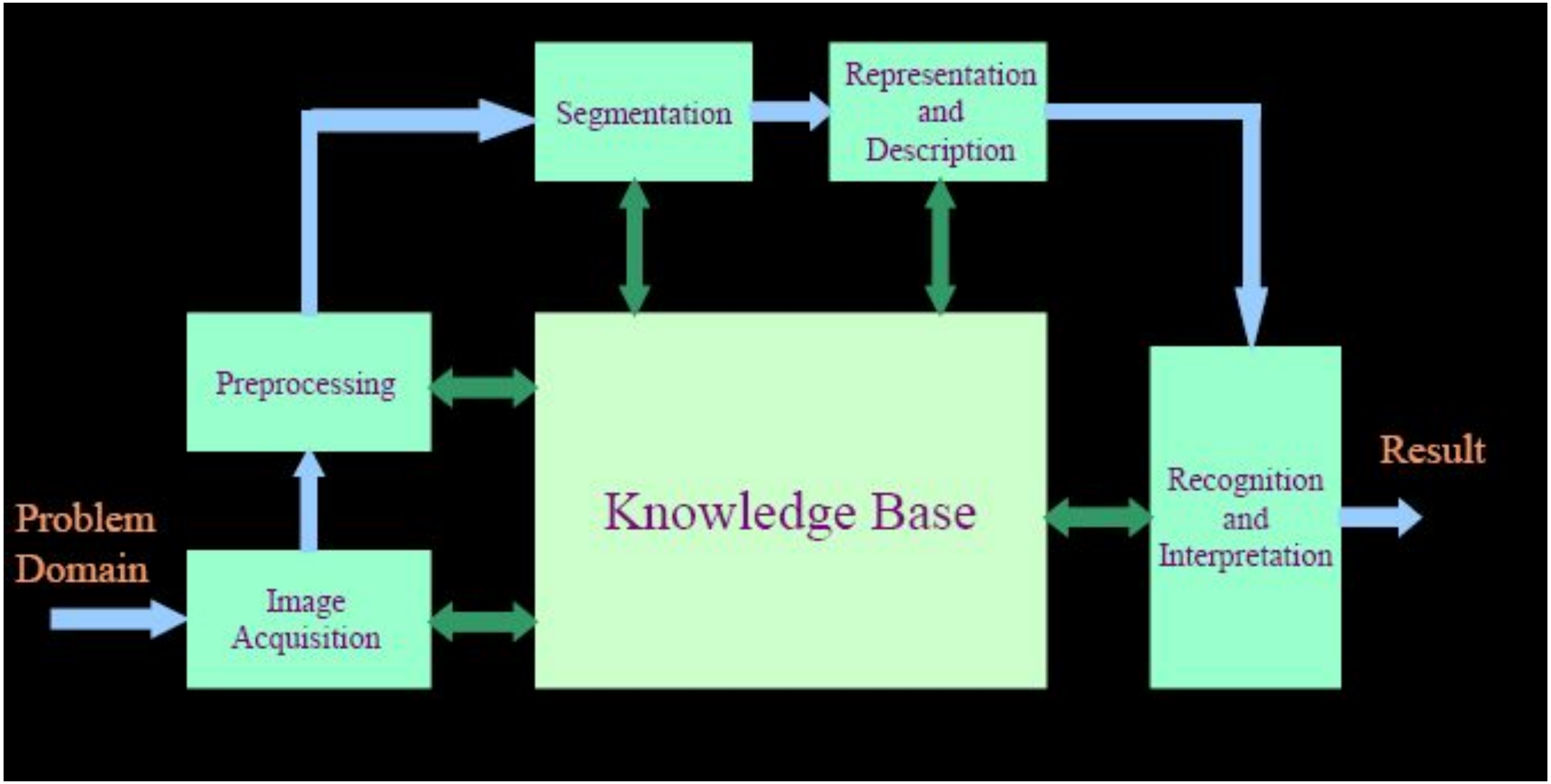
$$f(x, y) = [f_R(x, y), f_G(x, y), f_B(x, y)].$$

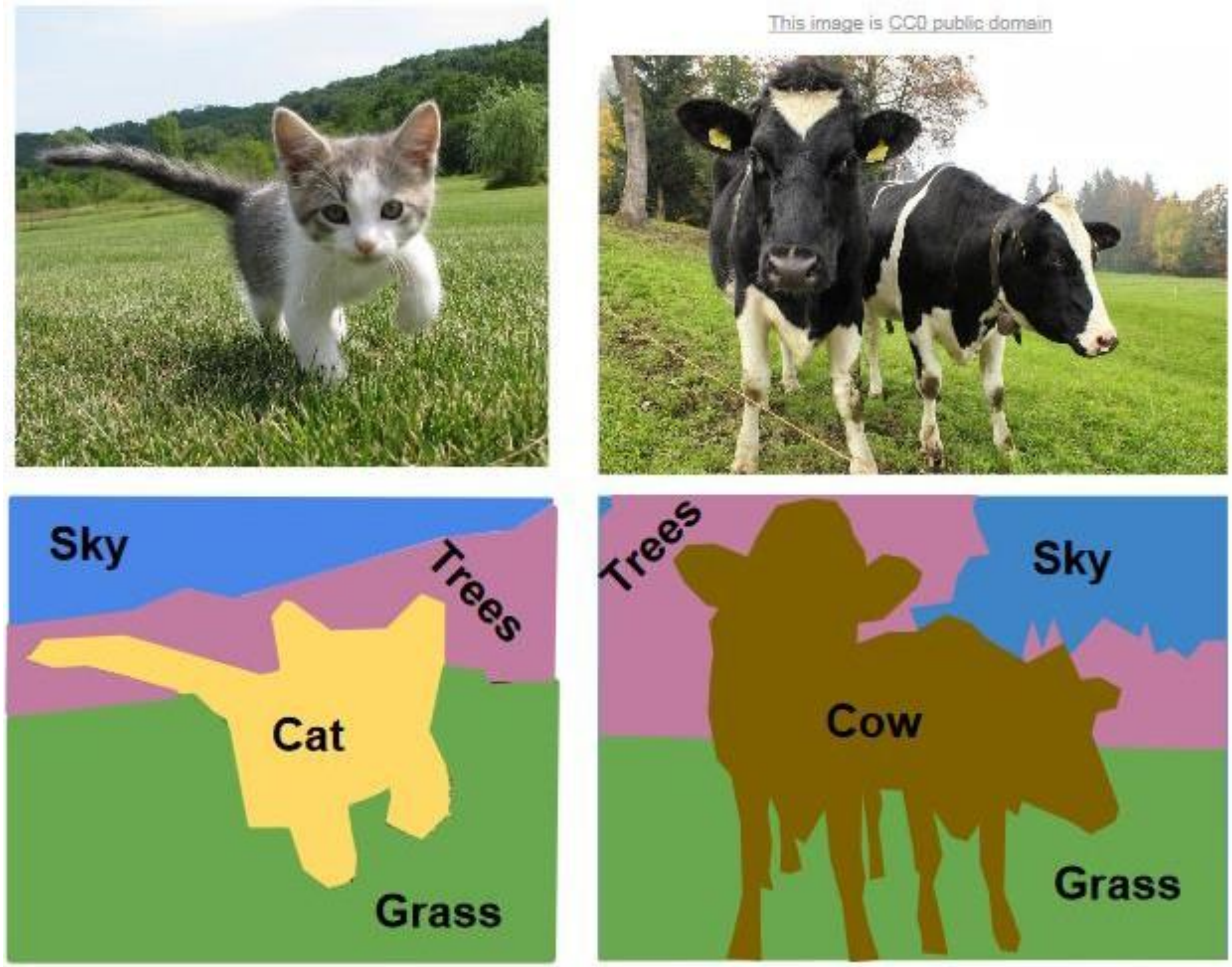
Steps in Digital Image Processing

Digital Image Processing involves following basic tasks

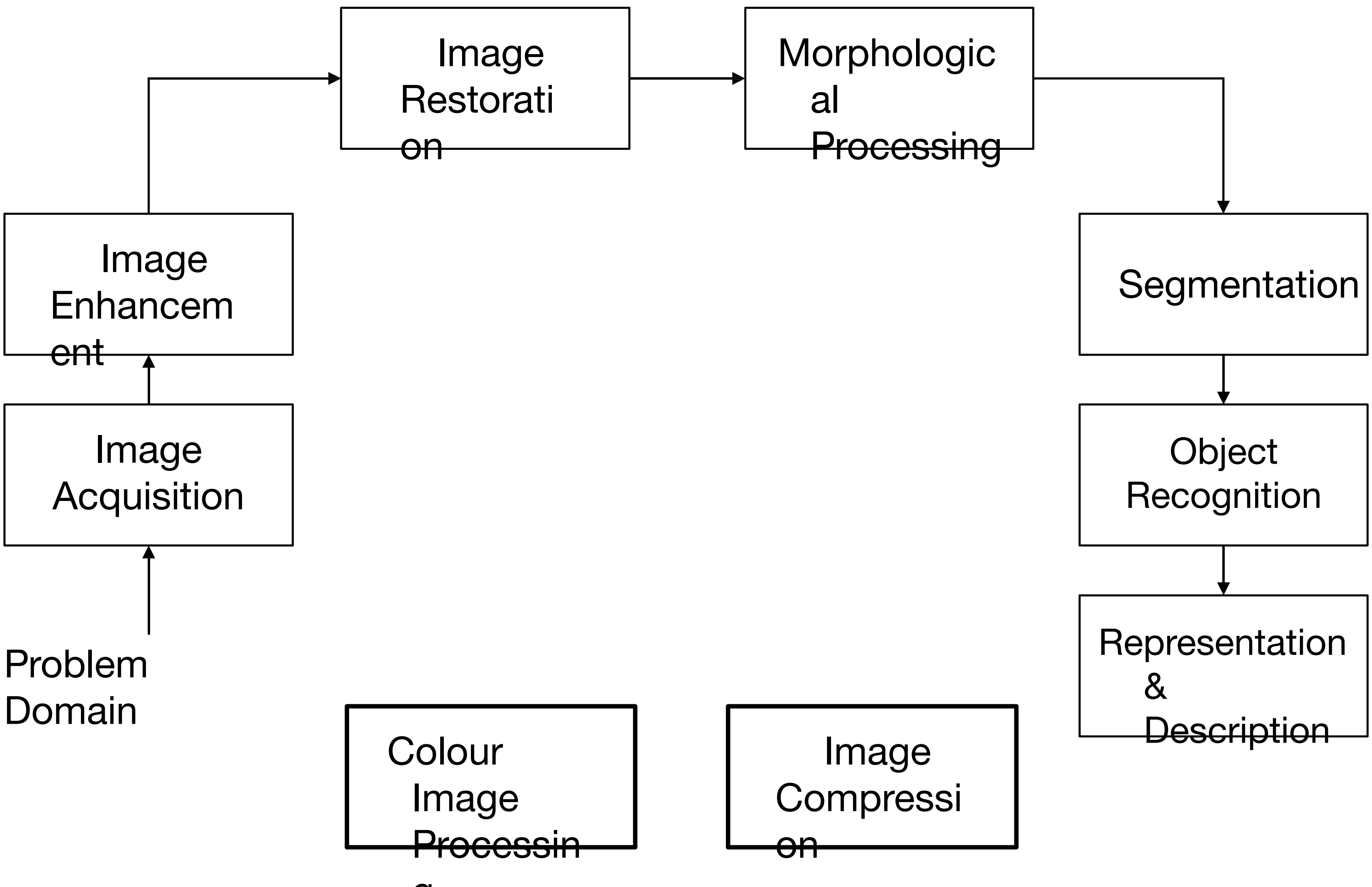
- **Image Acquisition:-** An imaging sensor and the capability to digitize the signal produced by the sensor
- **Preprocessing:-** Enhances theimage quality, filtering, contrast enhancement etc.
- **Segmentation:-** Partitions an input image into constituent parts of objects
- **Description/ Feature Selection:-** Extracts description of image objects suitable for further computer processing
- **Recognition & Interpretation:-** Assigning a label to the object based on the information provided by its descriptor. Interpretation assigns meaning to a set of

Steps in Digital Image Processing




This image is CC0 public domain
Sky
Trees
Cat
Grass
Trees
Sky
Cow
Grass

Key Stages in Digital Image Processing



Key Stages in Digital Image Processing: Image Aquisition

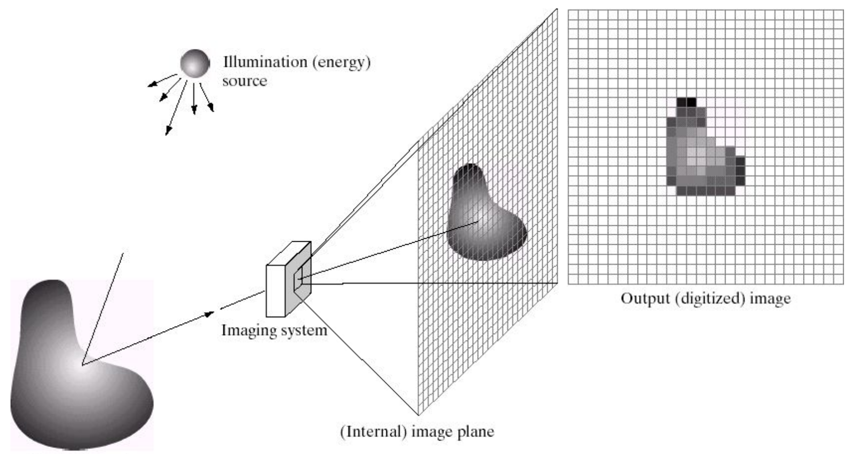
Image
Restorati
on

Morphologic
al
Processing

Image
Enhancem
ent

Segmentation

Image
Acquisition



Object
Recognition

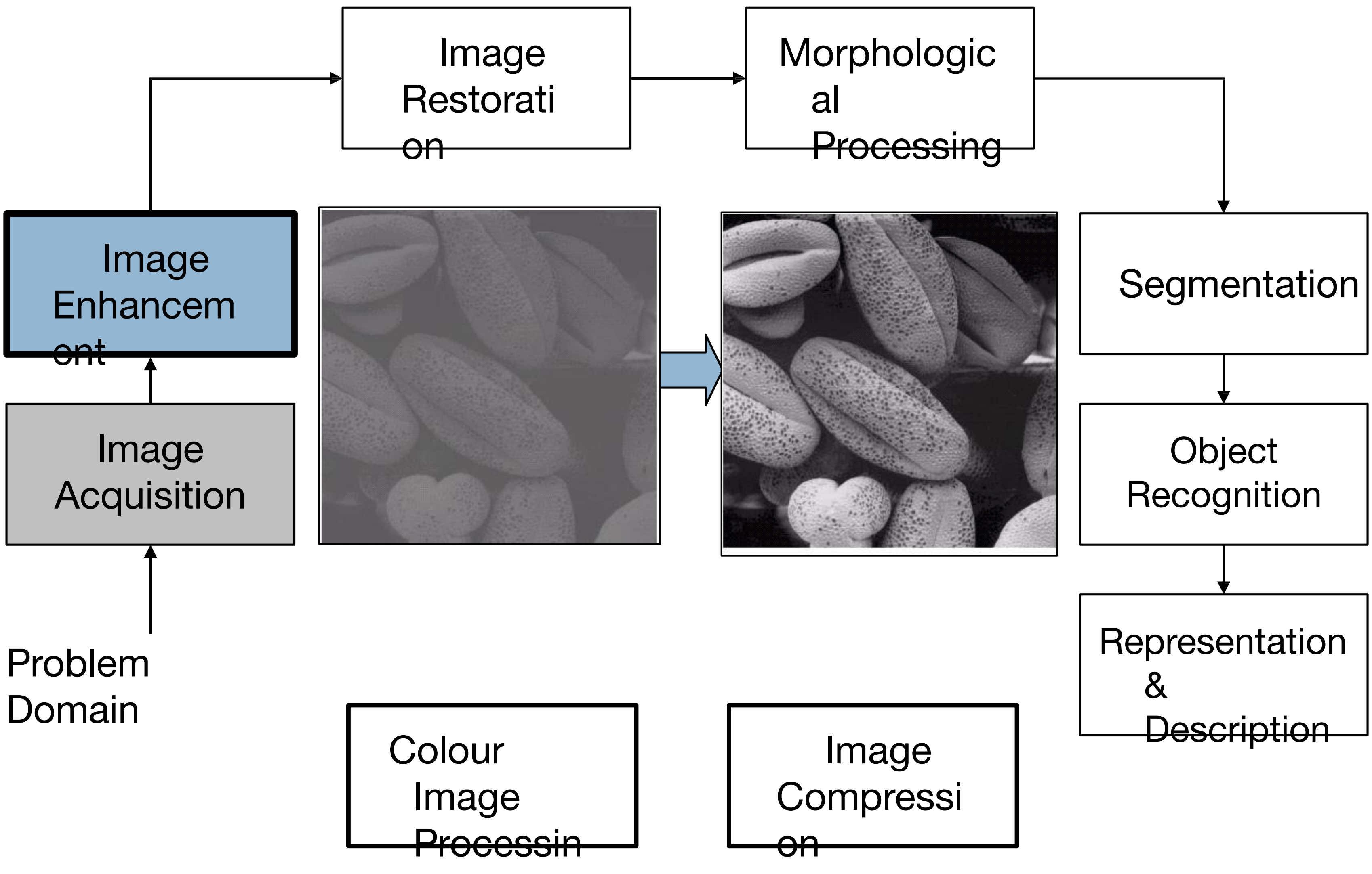
Representation
&
Description

Problem
Domain

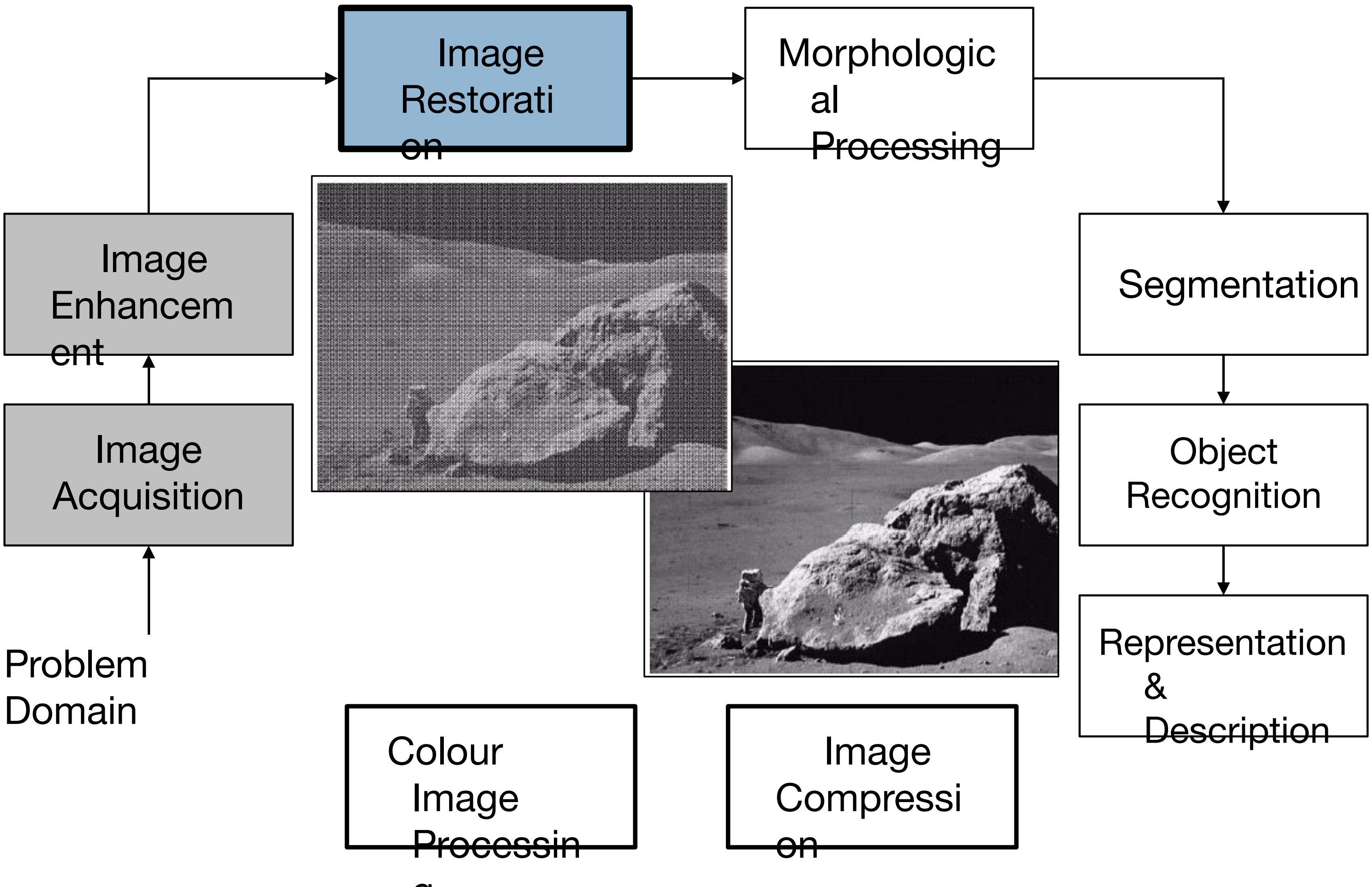
Colour
Image
Processin
g

Image
Compressi
on

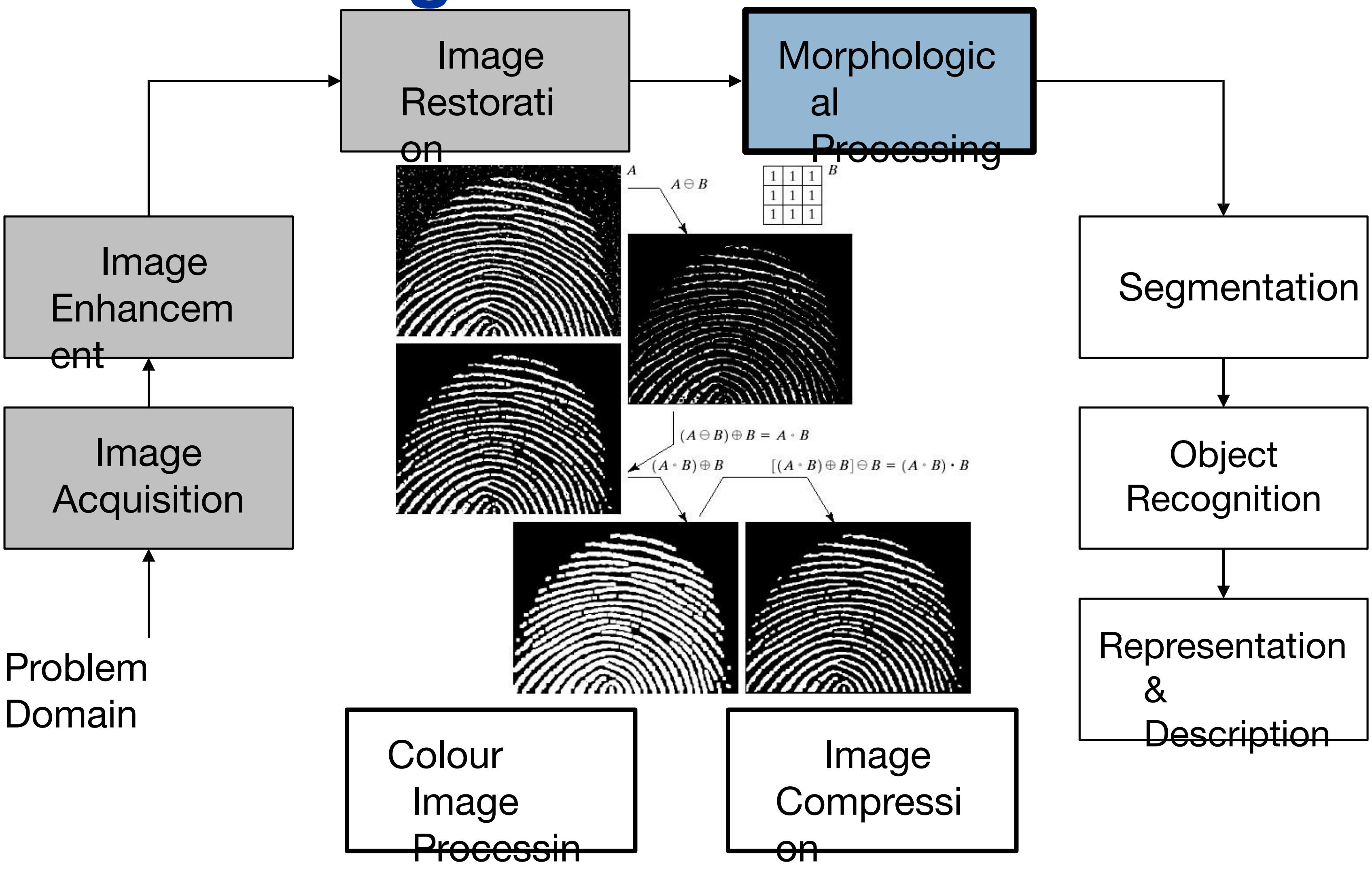
Key Stages in Digital Image Processing: Image Enhancement



Key Stages in Digital Image Processing: Image Restoration



Key Stages in Digital Image Processing: Morphological Processing



Key Stages in Digital Image Processing: Segmentation

Image Restoration

Morphological Processing

Image Enhancement

Segmentation

(a) (b)

Image Acquisition

Object Recognition

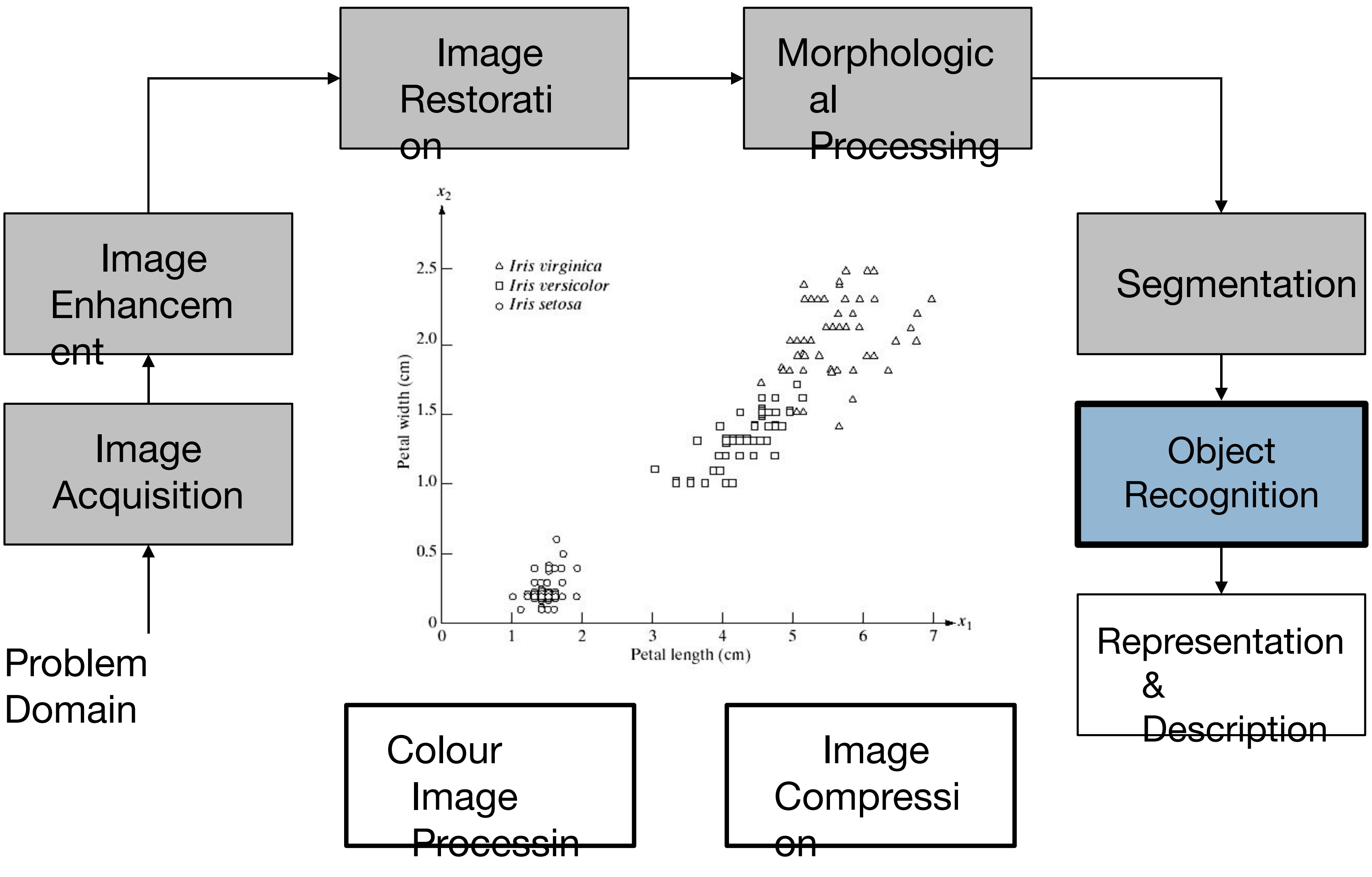
Representation & Description

Problem Domain

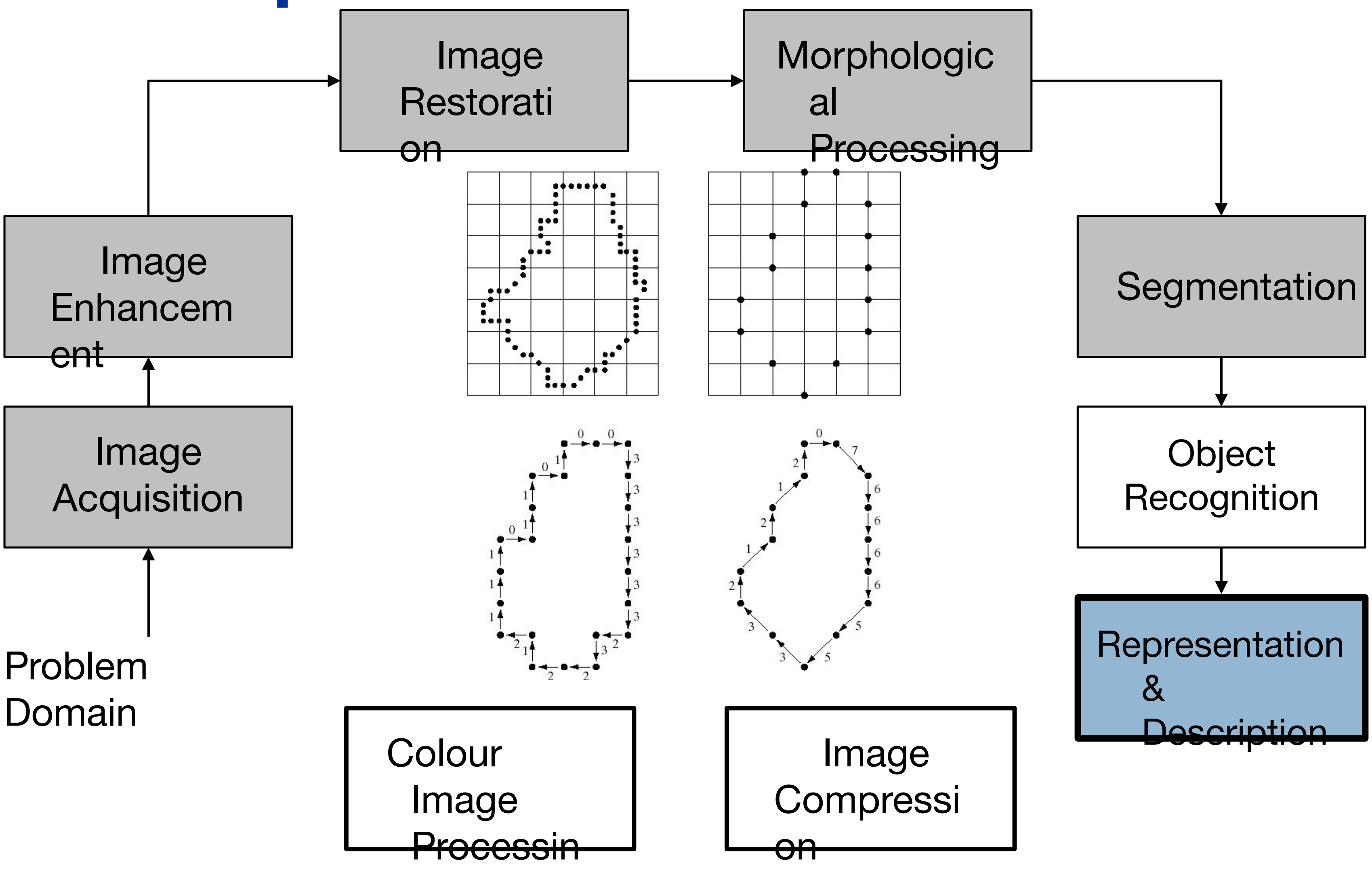
Colour Image Processing

Image Compression

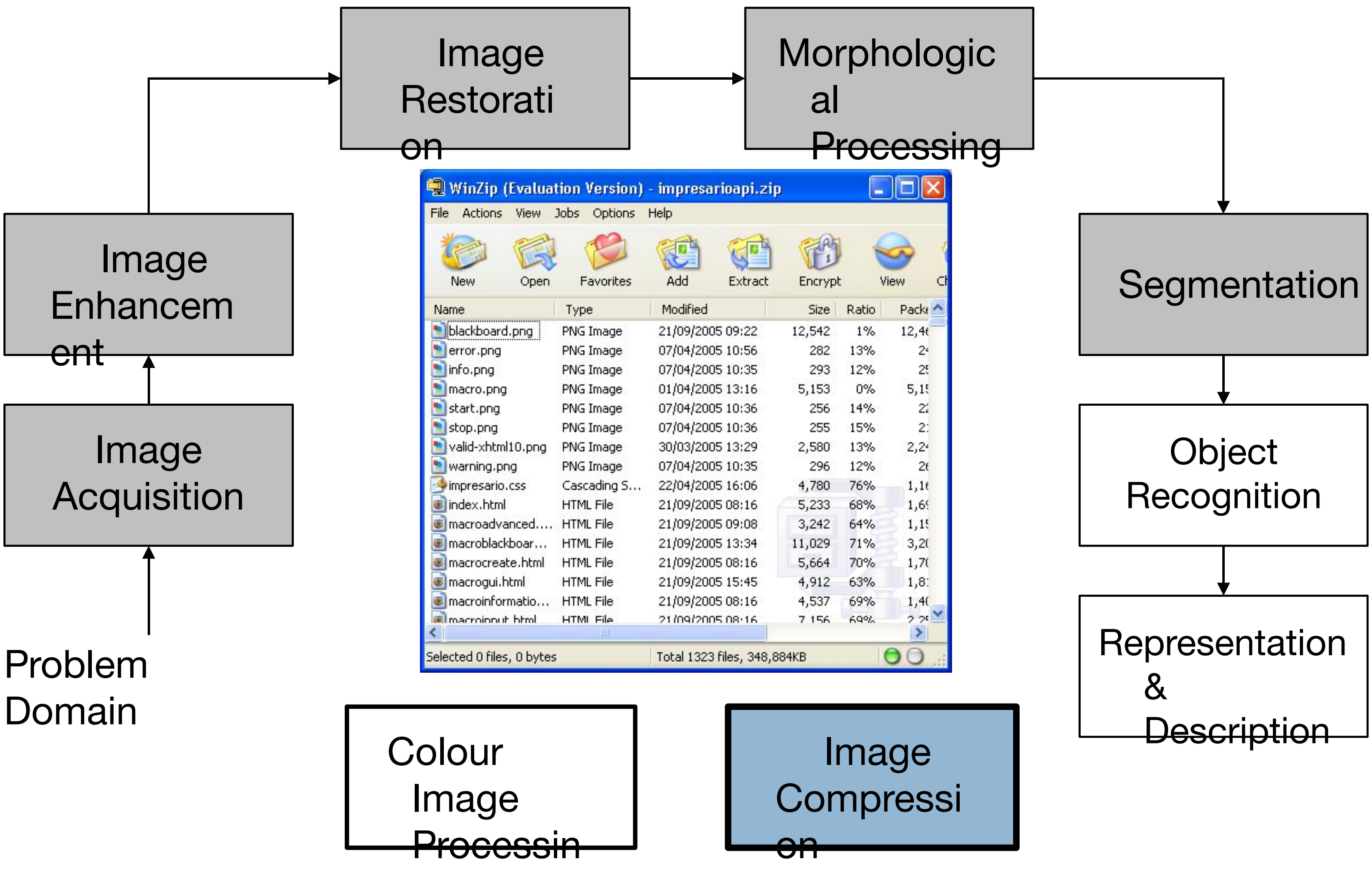
Key Stages in Digital Image Processing: Object Recognition



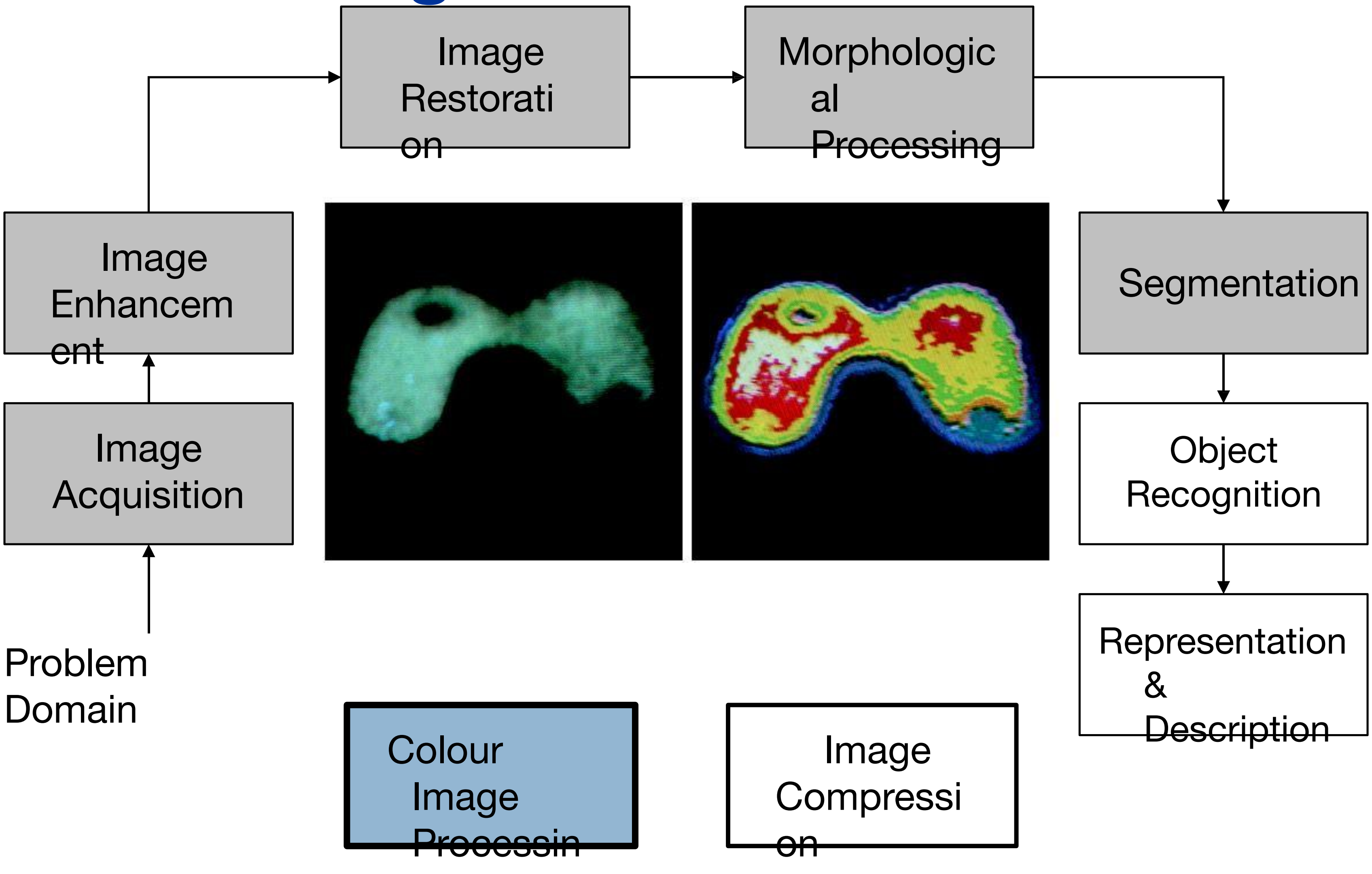
Key Stages in Digital Image Processing: Representation & Description



Key Stages in Digital Image Processing: Image Compression



Key Stages in Digital Image Processing: Colour Image Processing



Another Example

Image processing stages – acquisition

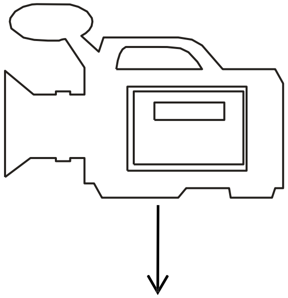


Image processing stages – filtering

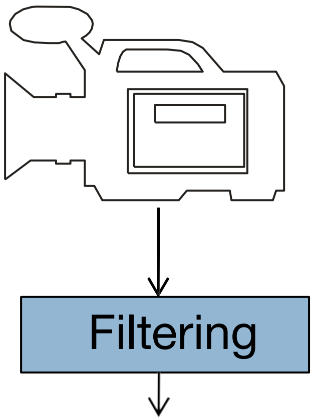
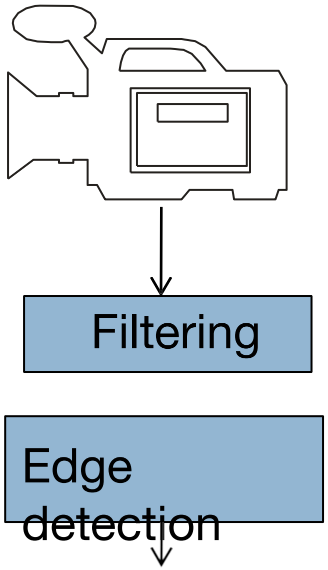


Image processing stages – edge detection



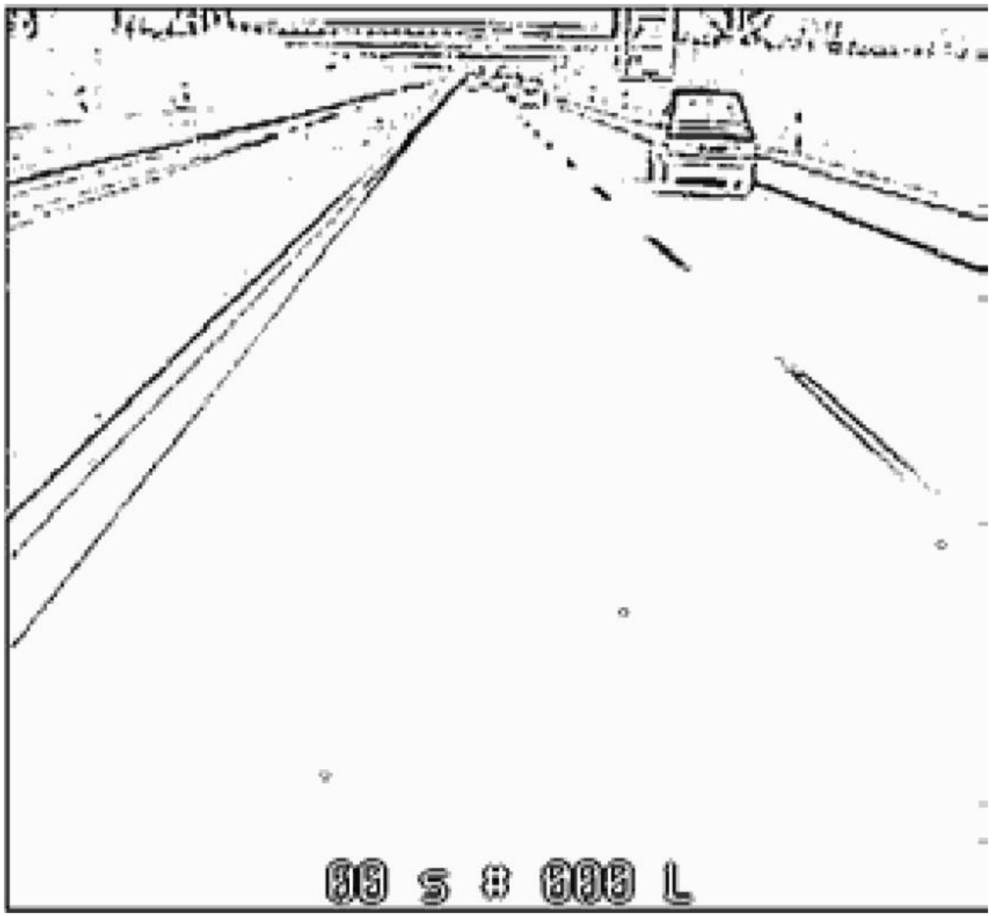
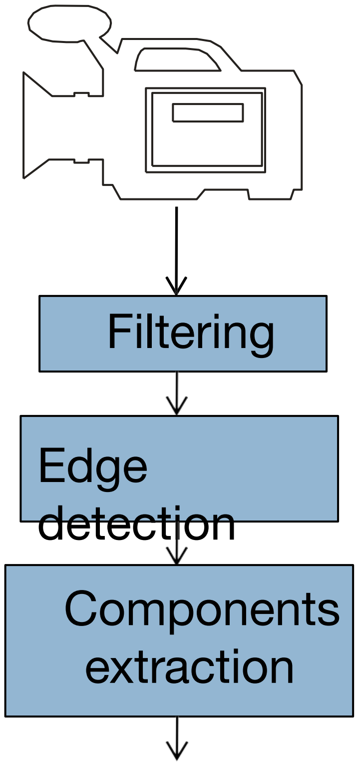


Image processing stages – components extraction



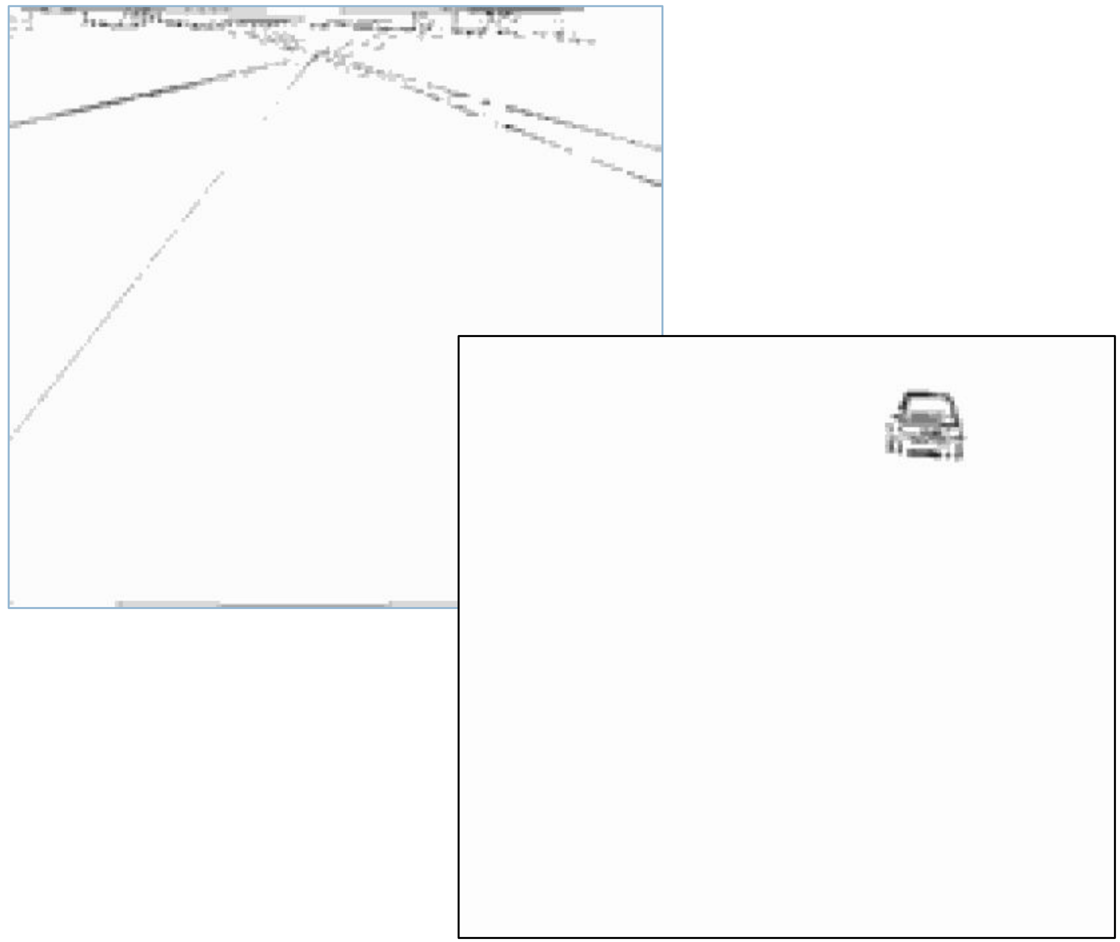
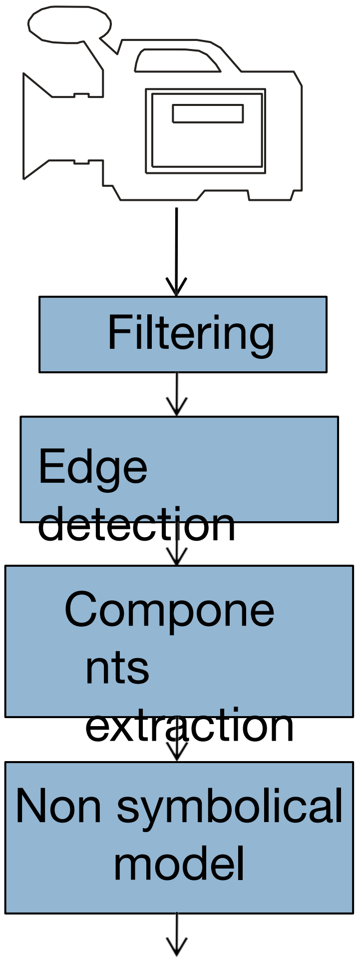
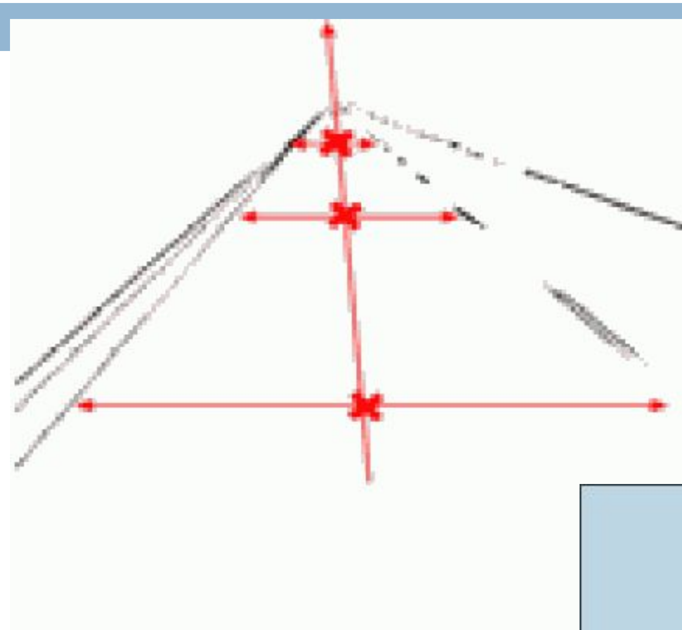
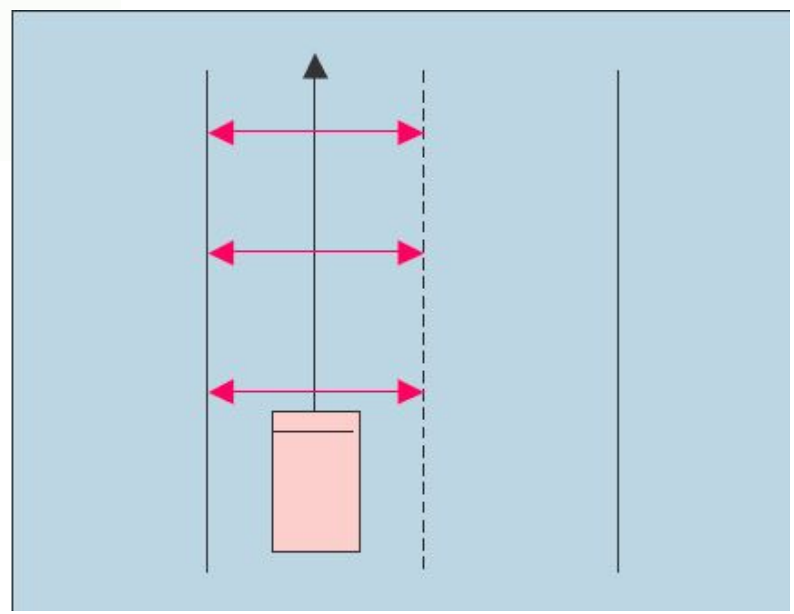


Image processing stages – non-symbolical



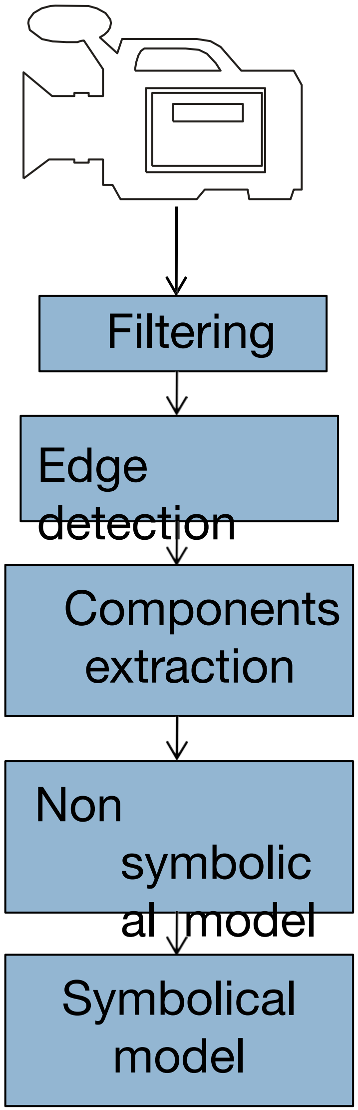


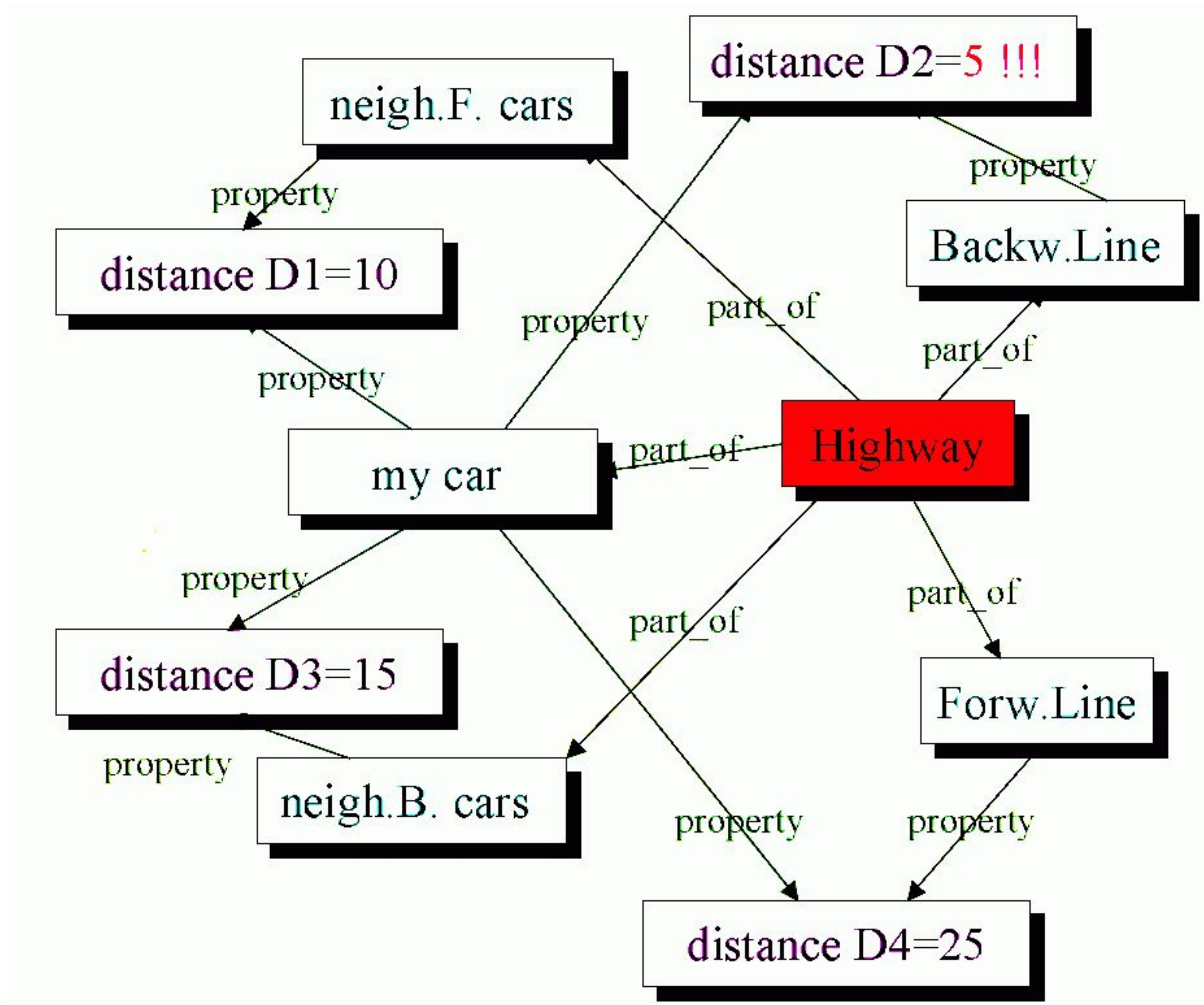
The image



Iconical mode

Image processing stages – symbolical model





Summar y

- We have looked at:
  - What is a digital image?
  - What is digital image processing?
  - History of digital image processing
  - State of the art examples of digital image processing
  - Key stages in digital image processing
- Next time we will start to see how it all works…

Any Questions?